PULLBACK FISHER–RAO GEOMETRIES OF PARAMETRIC PROBABILITY FAMILIES: AN EXPOSITORY ATLAS

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Abstract. We study the Fisher-Rao geometry induced on finite-dimensional statistical manifolds by pulling back the Fisher-Rao metric from the space of positive probability densities to the parameter space of a parametric probability model. Starting from this common construction, we derive and interpret the resulting Riemannian metrics for univariate and multivariate Gaussian distributions, elliptical Laplace distributions, linear and nonlinear Gibbs families, and probability families obtained as pushforwards of a fixed reference distribution. For each family, we compute the differential of the parameter-to-density map, derive the pullback metric tensor, and give a geometric interpretation of its components in terms of location, scale, anisotropy, orientation, energy fluctuations, or local mass compression. We then study the associated geodesics, using closed-form expressions when available and discrete Riemannian energy minimization otherwise. The purpose of this article is expository: to provide a unified geometric reading of several classical parametric probability models and to show how the same Fisher-Rao construction leads to distinct geometries on their parameter spaces. We finally contrast this viewpoint with Wasserstein-Otto geometry at the infinitesimal level.

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1. Introduction

Geometry is fascinating in that it shapes intuition by giving form to mathematical objects. By making them visual, it allows us to look at them, deform them, measure them, or even traverse them, and thus to think about them differently. It is through this geometric lens that we propose to study certain probability distributions: parametric families then become geometric objects in their own right, on which distances can be measured and geodesics followed.

Let $\mathcal{P}_+(E)$ denote the set of strictly positive probability densities on a vector space E, and let g^{FR} denote the Fisher–Rao form that will be written below. Pulling this form back along a family $\theta \mapsto p_\theta$ equips the parameter space with a Riemannian geometry.

Figure 1 gives a first glimpse. The top row shows three univariate densities, each corresponding to one point in a family. The bottom row shows the parameter plane: the colours and black curves measure a separation from the marked point. For the Gaussian family, this is the exact Fisher–Rao distance. For the two Gibbs families, it is the Riemannian length of the Euclidean chord, which is an upper bound on the geodesic distance.

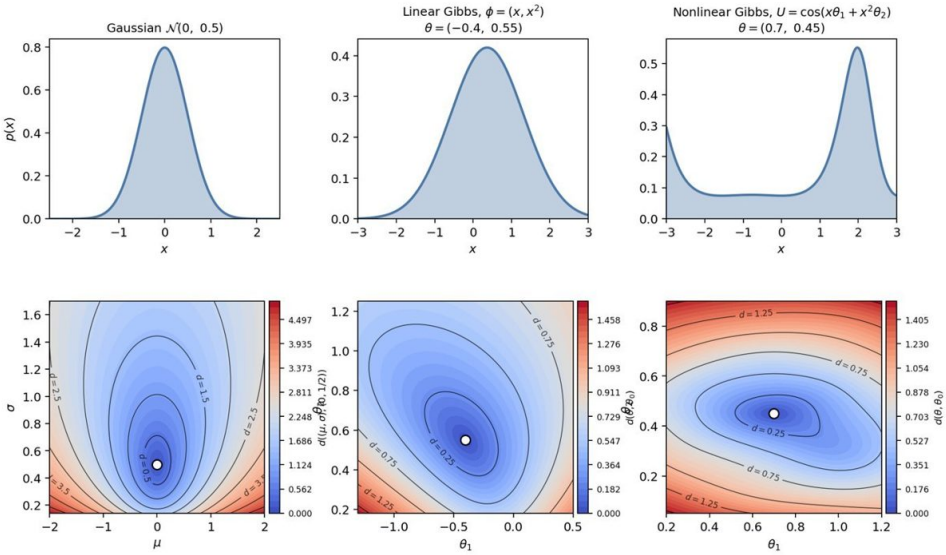


FIGURE 1. Three univariate parametric families (first row) and iso-contours in the parameter plane (second row). Left column: exact Fisher–Rao distance around $\mathcal{N}(0, 1/2)$. Middle and right columns: Riemannian length of the Euclidean chord (an upper bound on the geodesic distance), for the linear Gibbs family $\phi = (x, x^2)$ and the nonlinear Gibbs family $U = \cos(x\theta_1 + x^2\theta_2)$.

In this paper, we first revisit the construction of the Fisher–Rao metric and the principle of pullback, before presenting an atlas of the geometries obtained for several familiar parametric families.

For each family, the scheme is the same: we specify the parameterization, show a few members, compute the differential of $\theta \mapsto p_\theta$ into $\mathcal{P}_+(E)$, derive the metric from it, and then analyze this metric and consider several geodesics. A reader more interested in intuition may skim the calculations and focus on the form of the metric, its interpretation, and the figures showing the geodesics.

Most of the individual ingredients considered below are classical when taken separately. The Fisher information geometry of Gaussian models has been studied extensively [11, 12, 14, 15, 16], general elliptical families were treated, for instance, by Mitchell [13], and the Hessian and dualistic geometry of exponential families is standard in information geometry [17, 18]. The relation between densities and diffeomorphism groups underlying the push-forward viewpoint is also part of the broader geometric picture developed in [6]. The purpose of the present article is not to introduce a new information metric, but to place these constructions within a single pullback framework and to compare, through explicit calculations and visualizations, how different parametrizations turn the same ambient Fisher–Rao metric into different finite-dimensional geometries. Particular emphasis is placed

on the geometric interpretation of the metric components, on the comparison of the resulting geodesics, and on the push-forward formulation in terms of local mass compression.

Overview of the atlas

The same construction will be used throughout the paper, but the geometry induced on parameter space takes rather different forms from one family to another. Table 1 gives a compact roadmap. Its purpose is only to orient the reader; all formulas are derived and interpreted in the corresponding sections.

Family	Parameter space	Pullback Fisher–Rao metric	Geometric reading
Univariate Gaussian	$\mathbb{R} \times \mathbb{R}_+^*$	$ds^2 = \dfrac{d\mu^2 + 2\,d\sigma^2}{\sigma^2}$	Hyperbolic geometry; translation becomes cheaper as the distribution spreads. Geodesics are known in closed form.
Multivariate Gaussian	$\mathbb{R}^n \times \mathbb{S}_{++}^n$	$ds^2 = d\mu^\top \Sigma^{-1} d\mu + \frac{1}{2}\operatorname{tr}\big(\Sigma^{-1} d\Sigma \Sigma^{-1} d\Sigma\big)$	Mean displacement, change of scale and anisotropy, and rotation of the principal axes separate naturally.
Elliptical Laplace	$\mathbb{R}^n \times \mathbb{S}_{++}^n$	$ds^2 = a_n\, d\mu^\top \Sigma^{-1} d\mu + b_n\, \operatorname{tr}\big(\Sigma^{-1} d\Sigma \Sigma^{-1} d\Sigma\big) + c_n \operatorname{tr}\big(\Sigma^{-1} d\Sigma\big)^2$	The Gaussian decomposition persists, but the dispersion block couples the principal scales through the isotropic mode.
Linear Gibbs	$\Theta \subset \mathbb{R}^k$	$G(\theta) = \operatorname{Cov}_\theta\big(\phi(X)\big) = \operatorname{Hess}_\theta \log Z_\theta$	The metric measures fluctuations of the features $\phi(X)$ and is generated by a single convex potential.
Nonlinear Gibbs	$\Theta \subset \mathbb{R}^k$	$G(\theta) = \operatorname{Cov}_\theta\big(\nabla_\theta U(\theta, X)\big)$	The covariance interpretation remains, while the pure Hessian structure of the linear family is lost in general.
Pushforward	$\Theta \subset \mathbb{R}^k$	$g_\theta(d\theta_1, d\theta_2) = \mathbb{E}_\theta[\xi_{d\theta_1}\xi_{d\theta_2}]\,, \quad \xi_{d\theta} = \dfrac{\operatorname{div}(p_\theta v_{d\theta})}{p_\theta}$	Fisher–Rao measures the local compression or dilation of probability mass, rather than the magnitude of the underlying displacement field.

TABLE 1. Roadmap of the parametric families considered in the paper. The ambient Fisher–Rao construction is fixed; what changes is the way each parametrization deforms the probability density.

2. Framework and notation

It is not easy to decide what may be assumed known to the reader, nor to maintain a level of rigor that leaves room for intuition while remaining

formal enough to support the rest of the paper. This section gathers the notation and background that will be needed. It may be skimmed or skipped on a first reading, and consulted again whenever the paper refers back to it, or when a notation or concept is missing.

2.1. **Probability distributions**

We work on the measure space $(E, \mathcal{B}(E), \lambda)$, where E denotes a Euclidean space (most often $\mathbb{R}^n$), $\mathcal{B}(E)$ the Borel σ-algebra associated with the topology induced by the inner product on E, and λ the corresponding Lebesgue measure.

We consider probability measures defined on $\mathcal{B}(E)$ that admit a density. If P is a probability measure absolutely continuous with respect to λ, written $P \ll \lambda$, then the Radon–Nikodym theorem ensures the existence of a function $p = \frac{dP}{d\lambda}$ such that

$$P(A) = \int_A p(x)\, d\lambda(x), \qquad \forall A \in \mathcal{B}(E).$$

Finally, we restrict our attention to strictly positive densities, and denote by $\mathcal{P}_+(E)$ the set of all $p \in L^1(E)$ such that $p > 0$ almost everywhere and $\int_E p = 1$. This set is not open in $L^1(E)$: the positive cone of L^1 has empty interior, and the mass constraint places it in an affine hyperplane. We shall therefore not treat it as a Banach manifold. Infinite-dimensional manifold structures on spaces of probability measures can be constructed under stronger choices of topology and regularity; see, for instance, Pistone and Sempi [4]. In what follows, $\mathcal{P}_+(E)$ is only a space of densities on which we write formulas; the Riemannian manifolds in the strict sense will be the finite-dimensional parametric families, once the metric has been pulled back to Θ.

2.2. **Fréchet differential**

The Fréchet differential is the intrinsic object underlying partial derivatives and Jacobian matrices. Let $f\colon E \to F$ be a map between two normed vector spaces. Its differential at $a \in E$, when it exists, is denoted by $df(a)$ or df_a: it is an element of $\mathcal{L}(E, F)$ such that

$$f(a + h) = f(a) + df_a(h) + o(\|h\|).$$

Norms on E and F are required for the little-o term to make sense. The linear map df_a itself does not depend on a choice of bases: this is what is meant by its intrinsic character.

When E is finite-dimensional and (e_i) is a basis, every $x = \sum_i x_i e_i$ gives

$$df_a(x) = \sum_i x_i\, df_a(e_i) = \sum_i x_i \frac{\partial f}{\partial x_i}(a) :$$

the $df_a(e_i)$ are the partial derivatives. If F is also finite-dimensional and bases B_E of E and B_F of F have been fixed, the Jacobian matrix of f at a is precisely the matrix of the linear map df_a in these bases,

$$J = \mathrm{Mat}_{B_E, B_F}(df_a),$$

with entries $J_{ij} = \partial_j f_i(a)$. Finally, if $F = \mathbb{R}$ and E is equipped with an inner product, there exists a unique vector $\nabla f(a) \in E$, the gradient, such that $df_a(x) = \langle \nabla f(a), x \rangle$.

Figure 2 shows what the differential does on a nonlinear map of the plane. The map f bends the Cartesian grid, so the image of the parallelogram spanned by u and v is no longer a parallelogram. The differential df_a, on the other hand, is linear: it sends that parallelogram to a genuine parallelogram at $f(a)$, and $df_a(u + v) = df_a(u) + df_a(v)$ holds exactly. The gap between the two, visible as the dashed segments, is the remainder $o(\|h\|)$.

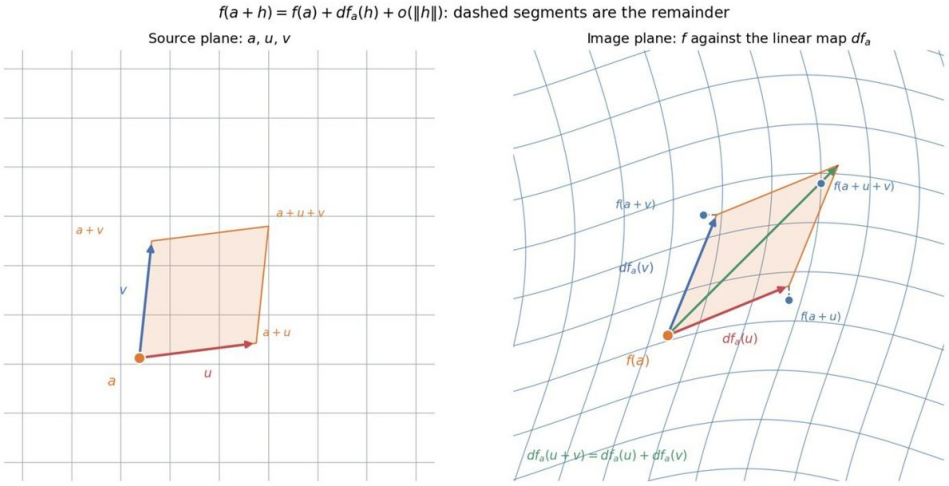


FIGURE 2. The Fréchet differential of $f(x, y) = (x + 0.35 \sin y,\ y + 0.35 \sin x)$ at the point a. Left: the Cartesian grid of the source plane, the base point a, the two vectors u and v, and the parallelogram they span. Right: the same grid transported by f, which curves it, together with the images $f(a + u)$, $f(a + v)$ and $f(a + u + v)$ of the three other vertices. Superimposed at $f(a)$ is the parallelogram generated by $df_a(u)$ and $df_a(v)$: the linear map df_a preserves the parallelogram structure, and its diagonal is $df_a(u + v) = df_a(u) + df_a(v)$. The dashed segments measure the distance between the linear model and the true images; they are the remainder $o(\|h\|)$, and they shrink faster than $\|h\|$ as u and v are shortened.

2.3. Riemannian manifold

2.3.1. Topological manifold

Recall that a *topological manifold* $\mathcal{M}$ of dimension k is a space equipped with a topology $\mathcal{T}$ and locally homeomorphic to $\mathbb{R}^k$, which can be written as follows:

$$\forall x \in \mathcal{M},\ \exists U \in \mathcal{T},\ \exists \varphi \colon U \to \varphi(U) \subset \mathbb{R}^k \text{ homeomorphism.}$$

An *atlas* of $\mathcal{M}$ is a collection $\mathcal{A} = \{(U_\alpha, \varphi_\alpha)\}$ of open sets in $\mathcal{T}$ and their associated homeomorphisms such that $\bigcup_\alpha U_\alpha = \mathcal{M}$.

The pair (U, φ) is a *local chart* of $\mathcal{M}$, and φ^{-1} is a *local parameterization*. A chart assigns coordinates to a point $x \in \mathcal{M}$ through the vector $(\pi_1(\varphi(x)), \dots, \pi_k(\varphi(x)))$. Figure 3 illustrates this idea: around a point p_0, the chart φ flattens a neighbourhood onto an open subset of $\mathbb{R}^k$; a second chart ψ, centred at another point p_1, does the same. On the overlap, the red grid can also be read in the plane of ψ: it is the image of $\varphi(U \cap V)$ under the transition map $\psi \circ \varphi^{-1}$, a $\mathcal{C}^\infty$ (and in particular $\mathcal{C}^1$) map from $\mathbb{R}^2$ to $\mathbb{R}^2$.

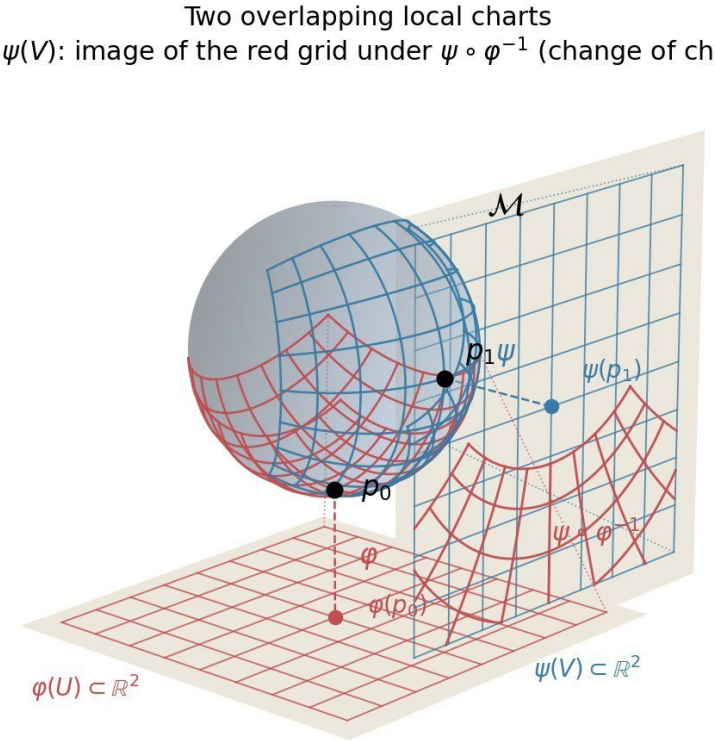


FIGURE 3. Two overlapping local charts. Here $\mathcal{M}$ is the sphere, p_0 its south pole, and p_1 its east pole. The chart φ (red) maps a neighbourhood of p_0 onto the lower plane; the chart ψ (blue) maps a neighbourhood of p_1 onto the plane on the right. The two grids, pushed onto $\mathcal{M}$ by the inverse maps, intersect. In the blue plane, the deformed red grid is the image of this overlap under $\psi \circ \varphi^{-1}$: the transition map remains smooth.

A manifold allows us to handle objects without first passing through coordinates. Two examples.

Neither a three-dimensional vector space such as $\mathbb{R}_2[X]$ (the polynomials of degree at most 2), nor the affine space of the room in which we are standing, is $\mathbb{R}^3$: these are objects, not lists of numbers. The polynomial

$-2 + X + 3X^2$ has no coordinates in itself, any more than a point in the room does. Once a basis of $\mathbb{R}_2[X]$ has been chosen, for example $(1, X, X^2)$, or an origin and a basis have been chosen in the ambient space, coordinates can be assigned. The polynomial becomes $(-2, 1, 3)$ through the map that sends P to its coefficients in the basis: this is a chart, and its inverse $(a, b, c) \mapsto a + bX + cX^2$ is a parameterization. A different choice of basis gives a different chart.

Another example is given by rotations of $\mathbb{R}^3$. Intrinsically, a rotation is an endomorphism R of $\mathbb{R}^3$ that preserves the inner product, $\langle R(x), R(y)\rangle = \langle x, y\rangle$, and the orientation, $\det R = 1$. The set of these maps is not a vector subspace of $\mathcal{L}(\mathbb{R}^3, \mathbb{R}^3)$, but a manifold. The axis–angle representation provides one chart, and Euler angles a second one. We pass from one to the other through the transition map, and thus obtain coordinates on rotations.

2.3.2. **Smooth manifold**

We say that $\mathcal{M}$ is a *smooth manifold* if there exists an atlas $\mathcal{A}$ such that, for every pair of charts $\big((U, \varphi), (V, \psi)\big) \in \mathcal{A}$ with $U \cap V \neq \emptyset$, the transition map $\varphi \circ \psi^{-1} \colon \mathbb{R}^k \to \mathbb{R}^k$ is of class $\mathcal{C}^\infty$.

This structure allows us to introduce the *tangent space* at a point $x \in \mathcal{M}$, denoted by $T_x\mathcal{M}$. It is constructed by considering equivalence classes of curves $\gamma \colon]-\epsilon, +\epsilon[\to \mathcal{M}$ satisfying $\gamma(0) = x$, according to their derivative at 0. For a chart defined in a neighbourhood of x, two curves are equivalent when

$$(\varphi \circ \gamma_1)'(0) = (\varphi \circ \gamma_2)'(0).$$

Here one may, if necessary, restrict the interval $I =]-\epsilon, +\epsilon[$ so that $\gamma_j(I) \subset U$ for $j \in \{1, 2\}$. The compositions are then well-defined maps

$$\varphi \circ \gamma_j \colon I \subset \mathbb{R} \longrightarrow \varphi(U) \subset \mathbb{R}^k,$$

and the derivatives above are ordinary derivatives of curves in $\mathbb{R}^k$. This definition does not depend on the chosen chart. The tangent space $T_x\mathcal{M}$ is the set of equivalence classes obtained in this way, and its elements are the *tangent vectors* to $\mathcal{M}$ at x.

Figure 4 makes this construction visible. Two curves γ_1 and γ_2 on $\mathcal{M}$ pass through the same point p with visibly different curvature, yet share a common velocity at p. In the chart φ their images are distinct curves through $\varphi(p)$, but they touch with the same derivative $v_\varphi = (\varphi\circ\gamma_1)'(0) = (\varphi\circ\gamma_2)'(0)$; the same holds in the chart ψ, with a common derivative v_ψ. The two chart representatives are related by the differential of the transition map: $v_\psi = d(\psi\circ\varphi^{-1})_{\varphi(p)}\, v_\varphi$. Thus $\gamma_1 \sim_p \gamma_2$, and the corresponding tangent vector does not depend on the chart.

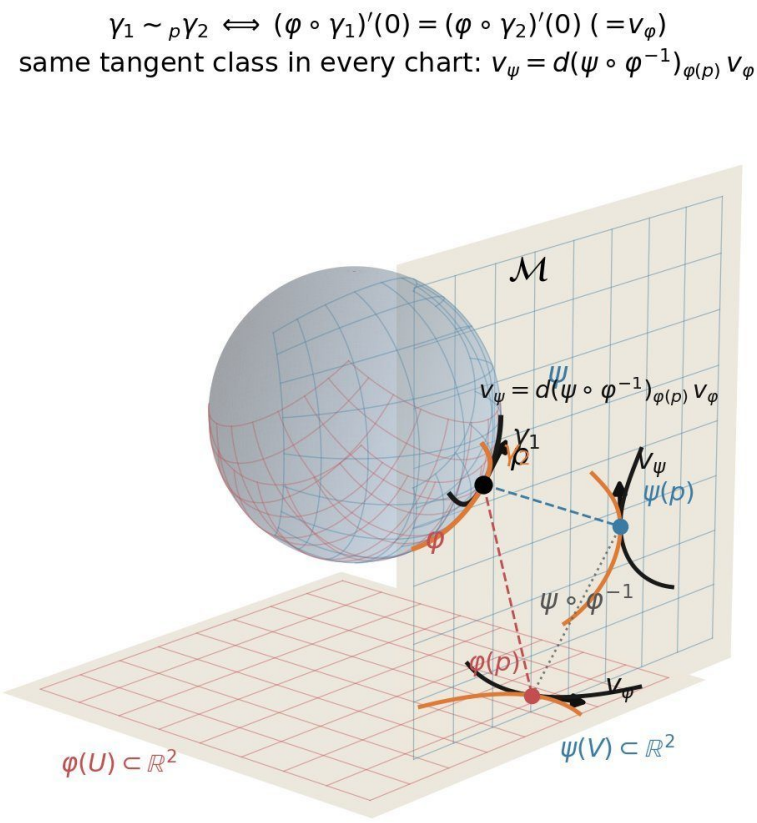


FIGURE 4. Intrinsic definition of a tangent vector as an equivalence class of curves. On the manifold $\mathcal{M}$, the curves γ_1 (black) and γ_2 (orange) both pass through p with the same velocity but different curvature. In each chart their images remain distinct, yet they share a common tangent arrow at the chart image of p: $v_\varphi = (\varphi \circ \gamma_1)'(0) = (\varphi \circ \gamma_2)'(0)$ and $v_\psi = (\psi \circ \gamma_1)'(0) = (\psi \circ \gamma_2)'(0)$. The transition map $\psi \circ \varphi^{-1}$ relates the two representatives by $v_\psi = d(\psi \circ \varphi^{-1})_{\varphi(p)}\, v_\varphi$, which is precisely the chart-independence of the equivalence relation $\gamma_1 \sim_p \gamma_2$.

If $\mathcal{M}$ is immersed in an ambient vector space $\mathbb{R}^n$ and a local parameterization $\phi\colon \mathbb{R}^k \to \mathcal{M} \subset \mathbb{R}^n$ is given around $x \in \mathcal{M}$, then

$$T_x\mathcal{M} \simeq \mathrm{Im}\big(d\phi_{\phi^{-1}(x)}\big),$$

where $d\phi_{\phi^{-1}(x)} \in \mathcal{L}(\mathbb{R}^k, \mathbb{R}^n)$ is the Fréchet differential of ϕ at the point $\phi^{-1}(x)$. A natural basis of $T_x\mathcal{M}$ is then $(\partial_{x_1}\phi, \dots, \partial_{x_k}\phi)$.

2.3.3. **Riemannian manifold**

Finally, we say that $\mathcal{M}$ is a *Riemannian manifold* if it is smooth and equipped with a $\mathcal{C}^\infty$ field of metric tensors g such that $g(x)\colon T_x\mathcal{M}\times T_x\mathcal{M}\to\mathbb{R}$ is a symmetric positive-definite bilinear form on the tangent space at x. When $T_x\mathcal{M}$ is equipped with a basis, we denote by $G(x)\in S^k_{++}(\mathbb{R})\subset M_k(\mathbb{R})$ the matrix of $g(x)$ in this basis.

In practice, g is written in three ways. For $u,v\in T_x\mathcal{M}$, the intrinsic form is $g_x(u,v)$. If a basis $B=(e_1,\dots,e_k)$ of $T_x\mathcal{M}$ is given and $U,V\in\mathbb{R}^k$ are the coordinates of u and v in B, then $g_x(u,v)=U^\top G(x)\,V$, where $G(x)$ is the matrix of g_x in B. In this basis, one also writes

$$ds^2=\sum_{i,j} g_{ij}(x)\,dx_i\,dx_j,$$

where $(dx_1,\dots,dx_k)$ is the dual basis of B, defined by $dx_i(e_j)=\delta_{ij}$. If $u=\sum_i U_ie_i$ and $v=\sum_j V_je_j$, we recover $ds^2(u,v)=\sum_{i,j} g_{ij}U_iV_j=U^\top GV$, the same formula as above.

The metric provides notions of angle between intersecting curves, curve length, distance between points, volume, and so on.

2.3.4. **Length, distance, geodesics**

The Riemannian metric g induces a notion of length for curves on $\mathcal{M}$. Let $\gamma\colon[a,b]\to\mathcal{M}$ be piecewise $\mathcal{C}^1$. Its length is

$$L(\gamma)=\int_a^b \sqrt{g_{\gamma(t)}\big(\gamma'(t),\,\gamma'(t)\big)}\,dt.$$

The Riemannian distance between two points is the infimum of these lengths,

$$d_g(x,y)=\inf_{\gamma} L(\gamma),$$

where the infimum is taken over all piecewise $\mathcal{C}^1$ curves joining x to y. When $\mathcal{M}$ is connected, this defines a genuine distance on $\mathcal{M}$. The topology induced by d_g coincides with the original topology of the manifold: the open d_g-balls form a basis for $\mathcal{T}$. In that sense d_g does not change the underlying space; it only measures it.

A *geodesic* is a constant-speed curve which is locally length-minimizing: around each of its points, a sufficiently short restriction realizes the Riemannian distance between its endpoints.

3. The Fisher–Rao metric

The name comes from two papers. Fisher [1] introduces the information carried by a parametric family in the form of an inner product on the scores $\partial_i \log p_\theta$. Rao [2] interprets this information as a Riemannian line element on the parameter space: this is the pullback that will be written below.

The distinguished role of this metric is also reflected in characterization results of encov type; see, for instance, Campbell [3].

The construction below embeds positive densities, viewed as elements of $L^1(E)$ (a vector space with no natural inner product), into $L^2(E)$ equipped with its Hilbert geometry $\langle f, g\rangle_{L^2} = \int_E f\, g$. The map is the square-root map $p \mapsto 2\sqrt{p}$: pulling back the inner product of L^2 equips the densities with a geometry that is no longer Euclidean. On a finite alphabet, this embedding is a $\mathcal{C}^\infty$ immersion and the construction is rigorous. When E is continuous, we retain the same formulas, restricting them to the directions for which they make sense.

3.1. Construction of the metric

When the base space has finite cardinality, $\mathcal{P}_+$ is the interior of the simplex, an open subset of the hyperplane $\sum_i p_i = 1$. The map $\psi\colon p \mapsto 2\sqrt{p}$ sends it to the positive orthant of the sphere of radius 2 in L^2, and g^{FR} is simply the round metric of the sphere pulled back to the simplex (Figure 5). Here ψ is $\mathcal{C}^\infty$, its differential is $d\psi_p(u) = u/\sqrt{p}$, and

$$g^{\mathrm{FR}}_p(u, v) = \langle d\psi_p(u),\, d\psi_p(v)\rangle_{L^2} = \sum_i \frac{u_i v_i}{p_i}.$$

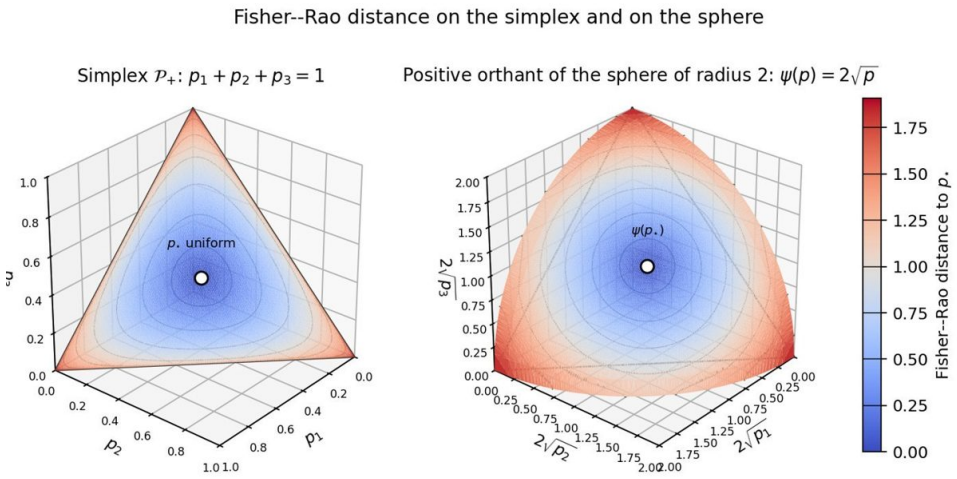


FIGURE 5. Effect of the embedding ψ on the metric. On the left, the simplex $\mathcal{P}_+$ of densities (p_1, p_2, p_3) with $p_1+p_2+p_3 = 1$; on the right, its image under $\psi : p \mapsto 2\sqrt{p}$, the positive orthant of the sphere of radius 2. Both surfaces are coloured by the Fisher–Rao distance to the uniform distribution $p_\star = (\frac{1}{3}, \frac{1}{3}, \frac{1}{3})$, marked in white, together with its isodistance curves. On the right, these are the geodesic circles of the sphere; on the left, their images under ψ^{-1}. These curves become denser near the boundary of the simplex: one of the coefficients p_i approaches 0, and the metric $u^\top \mathrm{diag}(1/p)\, u$ diverges.

In the continuous setting, $\psi(p) = 2\sqrt{p}$ still maps $\{p \in L^1(E) : p \geq 0\}$ into $L^2(E)$, since $\|\sqrt{p}\|_{L^2}^2 = \|p\|_{L^1}$. The map is continuous, but it is not Fréchet differentiable as a map $L^1 \to L^2$: $u \mapsto u/\sqrt{p}$ is not an element of $\mathcal{L}(L^1(E), L^2(E))$, and $u/\sqrt{p}$ belongs to L^2 only if $\int_E u^2/p < \infty$. We therefore do not claim to differentiate ψ on all of L^1. We retain the candidate

$$d\psi_p(u) = \frac{u}{\sqrt{p}},$$

defined precisely for those u for which this quotient is square-integrable, and set

$$g_p^{\mathrm{FR}}(u, v) = \int_E \frac{u(x)\, v(x)}{p(x)}\, dx$$

on this space. This is the same formula as in the finite case. The admissible directions at $p \in \mathcal{P}_+(E)$ are

$$T_p\mathcal{P}_+(E) = \Big\{u \in L^1(E) : \int_E u(x)\, dx = 0 \text{ and } \int_E \frac{u(x)^2}{p(x)}\, dx < \infty\Big\}.$$

The constraint $\int u = 0$ expresses conservation of mass to first order; $\int u^2/p < \infty$ is the condition for $d\psi_p(u)$ to exist in L^2, in other words, for $g_p^{\mathrm{FR}}(u, u)$ to be finite. Equipped with g_p^{FR}, this space is a Hilbert space. We denote it by $T_p\mathcal{P}_+(E)$ by analogy with a tangent space; it is not the tangent space of a Banach manifold structure on $\mathcal{P}_+(E)$.

3.2. Interpretation of the metric

For $u \in T_p\mathcal{P}_+(E)$, the norm is

$$\|u\|_{g_p^{\mathrm{FR}}}^2 = g_p^{\mathrm{FR}}(u, u) = \int_E \frac{u(x)^2}{p(x)}\, dx = \mathbb{E}_{x\sim p}\left[\left(\frac{u(x)}{p(x)}\right)^2\right].$$

In other words, what matters is not so much the absolute amplitude of $u(x)$ as the ratio $u(x)/p(x)$: the same change in density becomes more costly as $p(x)$ becomes smaller. Modifying the distribution where it is already rare has a high cost; modifying it where it is concentrated costs much less.

A numerical example makes this immediate. Consider a local increment $u(x) = 0{,}05$. If $p(x) = 0{,}1$, its contribution to the integrand is $u(x)^2/p(x) = 0{,}025$. If $p(x) = 100$, it falls to $2{,}5 \cdot 10^{-6}$. The absolute variation is the same, but the two costs are of entirely different orders.

This metric is also invariant under reparameterization of the observation space. If $\phi : E \to E'$ is a diffeomorphism and the densities are transported by ϕ, the inner products g^{FR} remain unchanged: only the way in which the densities differ from one another matters, not the coordinate system chosen on E. This invariance is in fact characteristic of the Fisher–Rao metric in several infinite-dimensional settings; see Bauer, Bruveris, and Michor [5].

It therefore differs from the Wasserstein metric, which arises from optimal transport [7]. The latter sees the points of E through a spatial cost $\|x - y\|$; moving mass along E has a cost that depends on the ambient geometry. Fisher–Rao, by contrast, measures a statistical difference: at the infinitesimal scale, the relative amplitude of density variations, that is, a local lack of overlap. Two distributions that are close in the Fisher–Rao sense may be

far apart in the Wasserstein sense, and conversely, as soon as the geometry of E comes into play.

Figure 6 considers again $p = \mathcal{N}(1,1)$. On the left, the same dipole u (with $\int u = 0$) is placed once in the bulk and once in the tail: $\int u^2/p$ increases from 0,008 to 0,16. On the right, p is translated; Wasserstein measures only the distance travelled by the mass, $W_2 = |\Delta\mu|$.

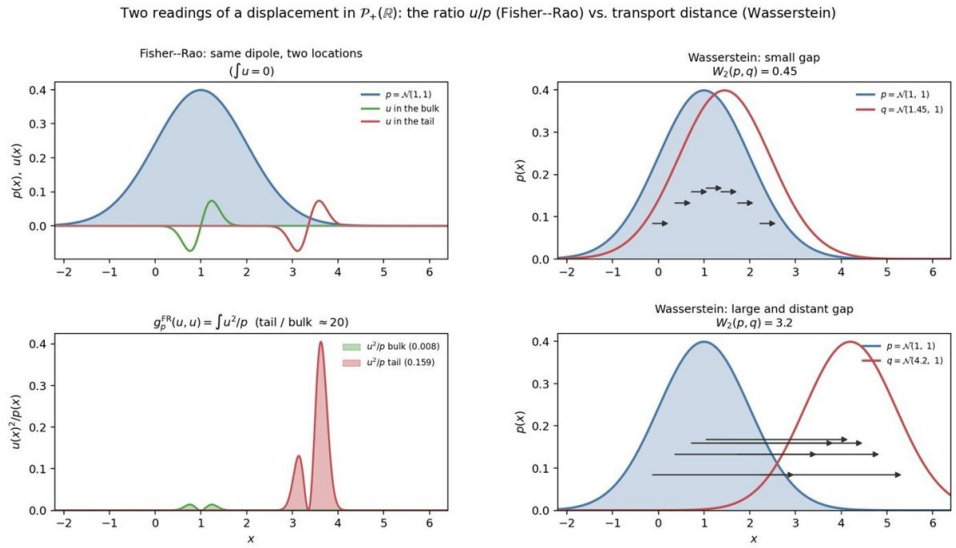


FIGURE 6. Two interpretations of a displacement in $\mathcal{P}_+(\mathbb{R})$. On the left, Fisher–Rao: the same dipole u, with $\int u = 0$, has a low cost at the centre of $p = \mathcal{N}(1,1)$ and a high cost in the tail, where p is small. On the right, Wasserstein: p is translated first by a small displacement and then by a large one; the cost is the length of the transport arrows.

3.3. **Pullback to the parameter space**

Consider a parametric family of probability densities taking values in $\mathcal{P}_+(E)$, represented by a function $p\colon \Theta \times E \to \mathbb{R}_+^*$, where $\Theta \subset \mathbb{R}^k$ is open and $p(\theta, x)$ denotes the density at the point $x \in E$ for the parameter θ. Depending on what we wish to emphasize, we write interchangeably $p(\theta, x)$, $p_\theta(x)$, or p_θ: the first form emphasizes the joint dependence on (θ, x), the second fixes the parameter and views the density as a function of x, and the last views it as a point of $\mathcal{P}_+(E)$. In the same spirit, the differential at θ is written $dp_\theta(v)(x)$, the value at x of the derivative of $\theta \mapsto p_\theta$ in the direction v.

We do not require the image of $\theta \mapsto p_\theta$ to be a submanifold of an infinite-dimensional manifold. We pull g^{FR} back to Θ, and it is (Θ, g), with $g = p_\theta^* g^{\mathrm{FR}}$, that forms the Riemannian manifold, provided that $G(\theta)$ below is well defined, of class $\mathcal{C}^\infty$, and positive definite. The injectivity of $dp_\theta\colon \mathbb{R}^k \to T_{p_\theta}\mathcal{P}_+(E)$ is precisely this positive definiteness: it is the immersion condition, expressed in the space of admissible directions.

The same operation is easier to read on a graph in $\mathbb{R}^3$. Let $\varphi(u, v) = (u, v, u^2 + v^2)$. The graph inherits the Euclidean metric of $\mathbb{R}^3$, which is pulled back to the (u, v) plane. Figure 7 colours both by the Riemannian distance to the point $\varphi(1, 1) = (1, 1, 2)$: the iso-contours on the graph are those of the chart. A small displacement ξ near $(1, 1)$ is mapped to $d\varphi(\xi)$, and

$$(\varphi^* \langle \cdot, \cdot \rangle)_{(1,1)}(\xi, \xi) = \left\langle d\varphi_{(1,1)}(\xi),\, d\varphi_{(1,1)}(\xi) \right\rangle .$$

This is the same operation as $g = p_\theta^* g^{\mathrm{FR}}$, with $\mathbb{R}^3$ in place of $\mathcal{P}_+(E)$ and φ in place of $\theta \mapsto p_\theta$.

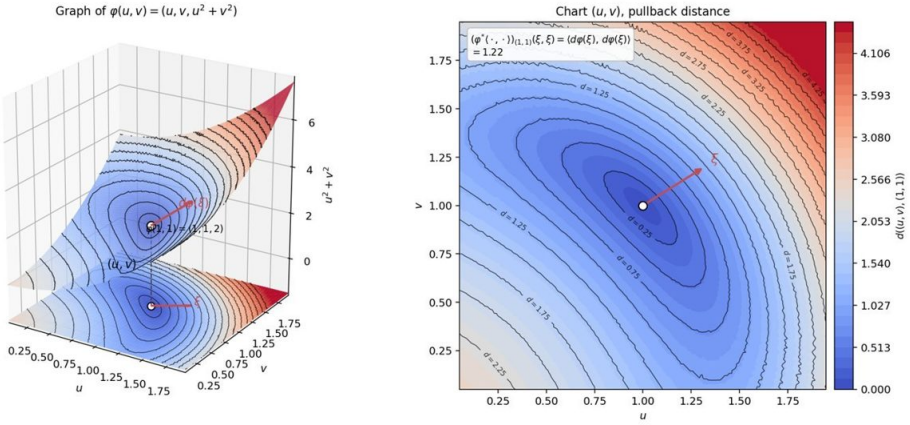


FIGURE 7. Graph of $\varphi(u, v) = (u, v, u^2 + v^2)$ in $\mathbb{R}^3$, coloured by a numerical approximation of the Riemannian distance to the point $(1, 1, 2)$ (geodesic shooting, Runge–Kutta 4), and the same function on the parameter plane (u, v). The iso-contours on the graph can be read again in the chart: this is the pullback. The arrow ξ in the plane is mapped to $d\varphi(\xi)$; by definition, $(\varphi^* \langle \cdot, \cdot \rangle)(\xi, \xi) = \langle d\varphi(\xi),\, d\varphi(\xi) \rangle$.

For every $\theta \in \Theta$, the tangent space $T_\theta \Theta \simeq \mathbb{R}^k$ is therefore equipped with the inner-product matrix

$$\boxed{G_{ij}(\theta) = \int_E \frac{\partial_{\theta_i} p(\theta, x)\, \partial_{\theta_j} p(\theta, x)}{p(\theta, x)}\, dx}$$

where $\dfrac{\partial p(\theta, x)}{\partial \theta_i} = d_\theta p(\theta)(e_i)(x)$. For the integral to be finite, we require $\partial_i p_\theta \in T_{p_\theta} \mathcal{P}_+(E)$, that is, $\int_E \partial_i p_\theta = 0$ and $\partial_i \log p_\theta \in L^2(p_\theta\, dx)$. The first identity follows by differentiating $\int_E p_\theta = 1$ under the integral sign, whenever a local domination $|\partial_i p_\theta| \le q_i \in L^1(E)$ permits it. The second condition is finite Fisher information. The families considered in the following sections satisfy it through an explicit calculation of the scores.

A curve $\theta(t)$ in Θ gives a curve of densities $p_{\theta(t)}$. Its length

$$L(\theta) = \int \sqrt{g_{\theta(t)}(\dot\theta(t), \dot\theta(t))}\, dt$$

is therefore the length of a path that remains within the parametric family $\{p_\theta\}$. The intrinsic distance on Θ is the infimum of these lengths over all such paths joining θ_0 to θ_1.

If the densities are viewed simply as elements of the ambient Hilbert space $L^2(E)$, the two endpoints can instead be joined by the straight line $(1-t)p_{\theta_0} + tp_{\theta_1}$. Its length is the ordinary L^2 distance

$$\left(\int_E \left(p_{\theta_0}(x) - p_{\theta_1}(x)\right)^2 dx\right)^{1/2},$$

which is not the Fisher–Rao distance. For the Fisher–Rao geometry, the embedding $p \mapsto 2\sqrt{p}$ identifies $\mathcal{P}_+(E)$ with the positive orthant of the sphere of radius 2 in $L^2(E)$. The corresponding ambient path is then an arc of a great circle on this sphere, and its length is

$$2\arccos\left(\int_E \sqrt{p_{\theta_0}(x)p_{\theta_1}(x)}\, dx\right).$$

This path may leave the parametric subfamily $\{p_\theta\}$. The distances considered in this paper are more constrained: their paths remain within Θ.

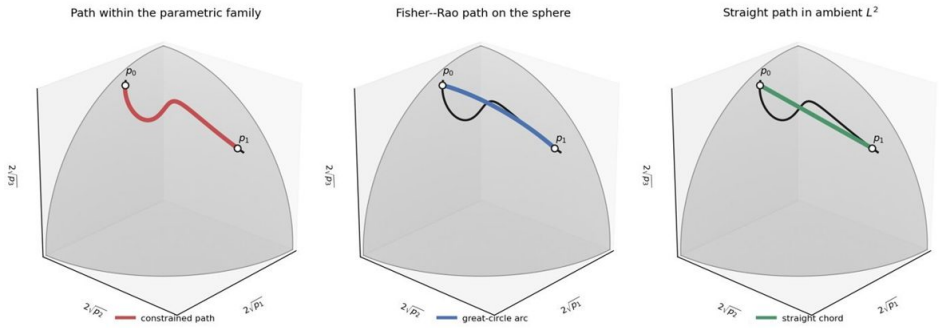


FIGURE 8. Three paths between the same two points under the square-root embedding $p \mapsto 2\sqrt{p}$. The grey surface is the positive orthant of the sphere of radius 2, and the black curve is a parametric subfamily. Left: the red path remains within the subfamily. Centre: the blue Fisher–Rao geodesic may leave the subfamily but remains on the sphere. Right: the green straight chord leaves the sphere and lies in the ambient L^2 space of square-root coordinates. This last path represents the chordal, or Hellinger, geometry and should not be confused with the ordinary L^2 straight line between the original densities.

3.4. A quick note on Natural gradient and steepest descent

There is one immediate consequence of equipping the parameter space with a metric. Let $f : \Theta \to \mathbb{R}$ be a cost function. As recalled in Section 2.2, its differential df_θ is a linear form on $T_\theta\Theta$. A gradient, however, is a vector, and turning the linear form df_θ into such a vector requires an inner product.

With the usual Euclidean structure on $\mathbb{R}^k$, the Euclidean gradient $\nabla_{\mathrm{euc}} f(\theta)$ is defined by

$$df_\theta(d\theta) = \langle \nabla_{\mathrm{euc}} f(\theta), d\theta \rangle .$$

It therefore reflects not only f, but also the Euclidean geometry chosen on the parameter coordinates.

Once Θ is equipped instead with the Riemannian metric g, the same differential has another representative. We denote by $\nabla_g f(\theta)$ the unique tangent vector such that

$$df_\theta(d\theta) = g_\theta(\nabla_g f(\theta), d\theta) , \qquad d\theta \in T_\theta\Theta.$$

Writing the metric in coordinates as

$$g_\theta(u, v) = u^\top G(\theta) v,$$

gives immediately

$$\nabla_g f(\theta) = G(\theta)^{-1} \nabla_{\mathrm{euc}} f(\theta).$$

When g is the Fisher–Rao metric pulled back from the space of densities, this Riemannian gradient is the *natural gradient* [22].

The factor $G(\theta)^{-1}$ has a simple geometric reading. Ordinary gradient descent regards all parameter directions as having the same local cost. The Riemannian metric replaces the Euclidean unit circle by

$$d\theta^\top G(\theta) d\theta = 1.$$

The direction of steepest descent is therefore no longer determined among Euclidean displacements of equal length, but among displacements of equal Riemannian length.

The natural gradient may therefore be viewed as the ordinary gradient after locally straightening the geometry of the parameter space. The function itself has not changed, but its graph expressed over these locally rescaled coordinates is stretched accordingly, and its direction of steepest variation changes with it.

A two-dimensional example makes the effect visible. Consider

$$f(\theta_1, \theta_2) = \frac{1}{2}\left(\theta_1^2 + \theta_2^2\right)$$

and the constant metric

$$G = \begin{pmatrix} 2 & 1 \\ 1 & 2 \end{pmatrix}.$$

The level sets of f are Euclidean circles, whereas the unit circles of g are tilted ellipses. At $\theta = (1, 0)$,

$$\nabla_{\mathrm{euc}} f = \begin{pmatrix} 1 \\ 0 \end{pmatrix}, \qquad \nabla_g f = G^{-1} \nabla_{\mathrm{euc}} f = \frac{1}{3} \begin{pmatrix} 2 \\ -1 \end{pmatrix}.$$

The two gradients therefore point in different directions. In coordinates that make the metric Euclidean, the second direction is again an ordinary steepest direction; its deviation in the θ-plane comes from expressing this geometry in the original parameters.

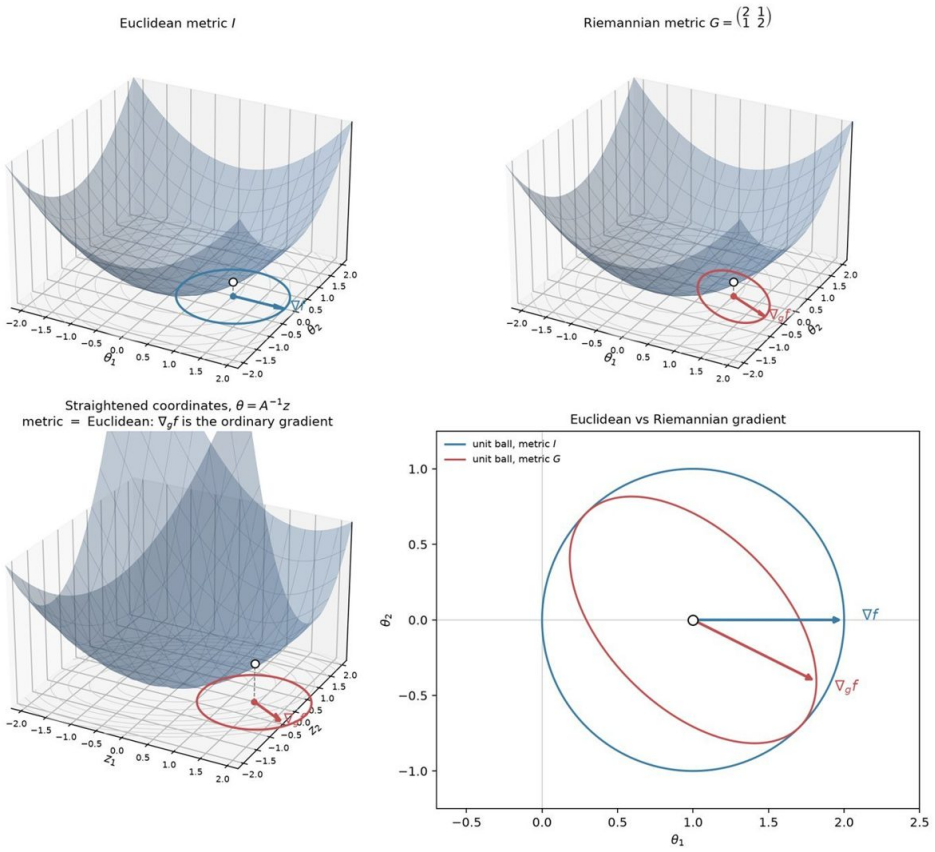


FIGURE 9. Geometric interpretation of the natural gradient in the two-dimensional example. Top left: the Euclidean unit circle and the Euclidean gradient on the graph of f. Top right: the same graph with the Riemannian unit ellipse $d\theta^\top G\, d\theta = 1$ and the Riemannian gradient. Bottom left: the coordinates $y = A(\theta - \theta_0)$, with $A = G^{1/2}$, straighten the metric locally and turn the ellipse into a Euclidean unit circle. Bottom right: direct comparison of the Euclidean and Riemannian gradients in the original parameter coordinates.

For a varying metric $G(\theta)$, this interpretation is local: the metric can always be made Euclidean at one point by such a linear change of tangent coordinates, but a single reparameterization need not flatten it everywhere. A first-order Riemannian gradient descent step in the present coordinates reads

$$\theta_{m+1} = \theta_m - \alpha\, G(\theta_m)^{-1} \nabla_{\mathrm{euc}} f(\theta_m).$$

For the families considered below, the matrix G is precisely the Fisher–Rao metric derived from the variation of the density. Natural gradient descent therefore measures a step in parameter space by the statistical change it produces, rather than by the Euclidean size of the parameter increment.

4. Gaussian family

Let us begin with the most emblematic family: the Gaussian family. It plays a central role for two reasons. For a fixed variance, it maximizes entropy [10], and it appears as the limiting distribution in the central limit theorem.

4.1. Univariate Gaussian

4.1.1. Differential

We start from the parametrization $p\colon \mathbb{R} \times \mathbb{R}_+^* \to \mathcal{P}_+(\mathbb{R})$ given by

$$p_{\mu,\sigma}(x) = \frac{1}{\sqrt{2\pi}\,\sigma}\, e^{-\frac{(x-\mu)^2}{2\sigma^2}}.$$

Figure 10 shows three members of this family, all centred at $\mu = 0$, with different standard deviations. This is the family being parametrized, viewed from the density side.

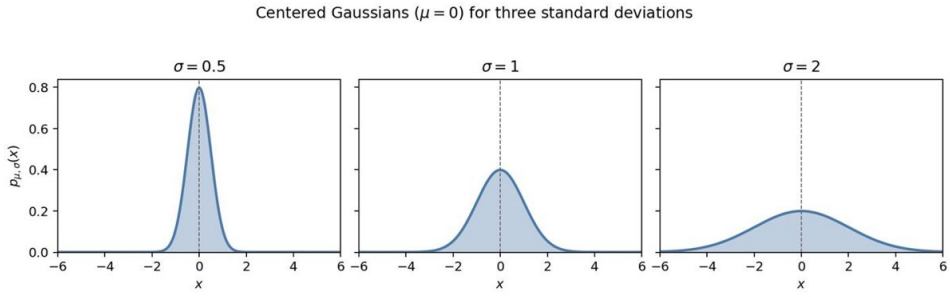


FIGURE 10. Three centred univariate Gaussians ($\mu = 0$), for $\sigma \in \{0{,}5,\ 1,\ 2\}$. The standard deviation spreads the density and lowers its peak.

Its partial derivatives with respect to μ and σ follow directly from $p_{\mu,\sigma}$:

$$\partial_\mu p_{\mu,\sigma}(x) = \frac{x-\mu}{\sigma^2}\, p_{\mu,\sigma}(x), \qquad \partial_\sigma p_{\mu,\sigma}(x) = \left[\frac{(x-\mu)^2}{\sigma^3} - \frac{1}{\sigma}\right] p_{\mu,\sigma}(x).$$

4.1.2. Metric

The pullback formula can be rewritten using the score $\partial_i \log p_{\mu,\sigma} = \partial_i p_{\mu,\sigma} / p_{\mu,\sigma}$. Indeed,

$$g_{ij}(\mu,\sigma) = \int_{\mathbb{R}} \frac{\partial_i p_{\mu,\sigma}\, \partial_j p_{\mu,\sigma}}{p_{\mu,\sigma}}\, dx = \mathbb{E}_{\mu,\sigma}\left[\partial_i \log p_{\mu,\sigma}(X)\, \partial_j \log p_{\mu,\sigma}(X)\right],$$

where $\mathbb{E}_{\mu,\sigma}$ denotes expectation under $p_{\mu,\sigma}$. This is the Fisher information, interpreted as a Riemannian metric by Rao [2]. The calculations then rely on the centred moments of the Gaussian, with $X \sim p_{\mu,\sigma}$: $\mathbb{E}[(X-\mu)^2] = \sigma^2$, $\mathbb{E}[(X-\mu)^3] = 0$, and $\mathbb{E}[(X-\mu)^4] = 3\sigma^4$.

For the $\mu\mu$ component,

$$g_{\mu\mu} = \mathbb{E}\left[\frac{(X-\mu)^2}{\sigma^4}\right] = \frac{\sigma^2}{\sigma^4} = \frac{1}{\sigma^2}.$$

The cross term involves only centred moments of odd order and therefore vanishes:

$$g_{\mu\sigma} = \mathbb{E}\left[\frac{X-\mu}{\sigma^2}\left(\frac{(X-\mu)^2}{\sigma^3} - \frac{1}{\sigma}\right)\right] = \frac{\mathbb{E}[(X-\mu)^3]}{\sigma^5} - \frac{\mathbb{E}[X-\mu]}{\sigma^3} = 0.$$

Finally, for the component associated with the standard deviation,

$$\begin{aligned} g_{\sigma\sigma} &= \mathbb{E}\left[\left(\frac{(X-\mu)^2}{\sigma^3} - \frac{1}{\sigma}\right)^2\right] = \frac{\mathbb{E}[(X-\mu)^4]}{\sigma^6} - 2\,\frac{\mathbb{E}[(X-\mu)^2]}{\sigma^4} + \frac{1}{\sigma^2} \\ &= \frac{3\sigma^4}{\sigma^6} - \frac{2\sigma^2}{\sigma^4} + \frac{1}{\sigma^2} = \frac{2}{\sigma^2}. \end{aligned}$$

The Fisher–Rao metric pulled back to the parameter space is therefore

$$g_{\mu,\sigma} = \begin{pmatrix} \dfrac{1}{\sigma^2} & 0 \\ 0 & \dfrac{2}{\sigma^2} \end{pmatrix}, \qquad \text{that is} \qquad ds^2 = \frac{d\mu^2 + 2\,d\sigma^2}{\sigma^2}.$$

4.1.3. **Analysis of the metric**

Up to a scaling factor, we recognize the metric of the Poincaré hyperbolic half-plane. More precisely, the change of coordinates $(\xi, \eta) = (\mu/\sqrt{2},\ \sigma)$ transforms the line element into

$$ds^2 = 2\,\frac{d\xi^2 + d\eta^2}{\eta^2},$$

which is twice the Poincaré metric $g_P = (d\xi^2 + d\eta^2)/\eta^2$ on the upper half-plane $\{(\xi, \eta) : \eta > 0\}$. The distances are therefore the Poincaré distances multiplied by $\sqrt{2}$. This identification is classical; it already appears in Atkinson and Mitchell [11], and Costa, Santos, and Strapasson [14] give a detailed geometric interpretation.

The associated Riemannian distance then has a closed-form expression. Between two points (μ_1, σ_1) and (μ_2, σ_2),

$$d\big((\mu_1,\sigma_1),(\mu_2,\sigma_2)\big) = \sqrt{2}\,\operatorname{arcosh}\left(\frac{(\mu_1-\mu_2)^2 + 2(\sigma_1^2+\sigma_2^2)}{4\,\sigma_1\sigma_2}\right).$$

Figure 11 shows this distance from the standard Gaussian $(\mu, \sigma) = (0, 1)$. The colours range from blue (near) to red (far); the black curves are level sets of the distance, in other words the circles of hyperbolic geometry in these coordinates. Approaching $\sigma = 0$ is clearly costly: the metric blows up near the boundary of the half-plane.

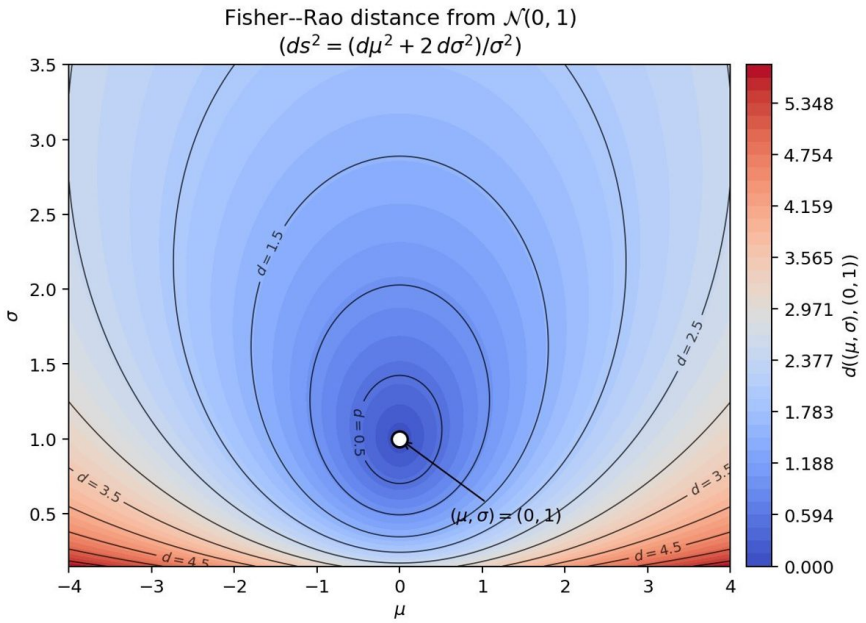


FIGURE 11. Fisher–Rao distance from $\mathcal{N}(0,1)$ in the parameter plane (μ,σ). Blue-to-red colour map and level sets; the "circles" of the metric are not Euclidean circles.

Here we recover, on the parameter space, the interpretation from section 3.2. The metric is governed by the variance. Moving from $\mathcal{N}(0,\ 0{,}1)$ to $\mathcal{N}(0{,}05,\ 0{,}1)$ is significantly more costly than the same shift in the mean under a larger variance, from $\mathcal{N}(0,\ 1)$ to $\mathcal{N}(0{,}05,\ 1)$. To first order, this is again a matter of overlap: at low variance, a small shift in μ already separates the two densities considerably; at high variance, the same shift leaves them largely superimposed (figure 12). The same phenomenon holds for variations in σ (figure 13). The geodesic distance minimizes length over paths $(\mu(t),\sigma(t))$ in the half-plane, which explains the detour: spreading the Gaussian has a cost, but it subsequently makes shifts in μ less costly.

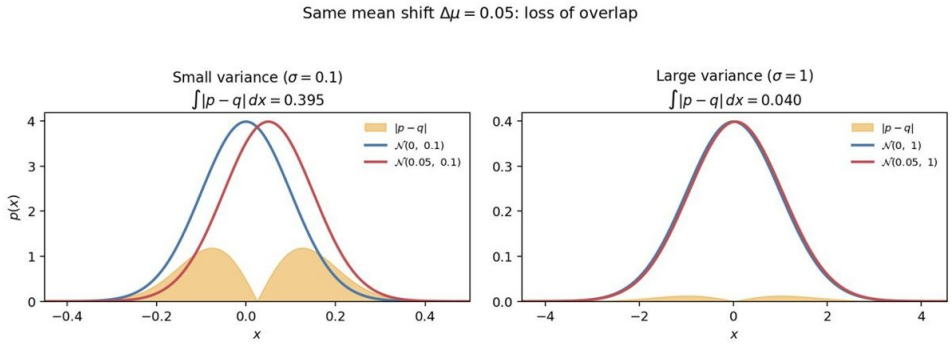


FIGURE 12. The same mean shift $\Delta\mu = 0{,}05$. On the left, with $\sigma = 0{,}1$, the densities separate clearly ($\int |p - q|\, dx \approx 0{,}40$); on the right, with $\sigma = 1$, they remain almost indistinguishable ($\int |p - q|\, dx \approx 0{,}04$). The orange area marks the loss of overlap.

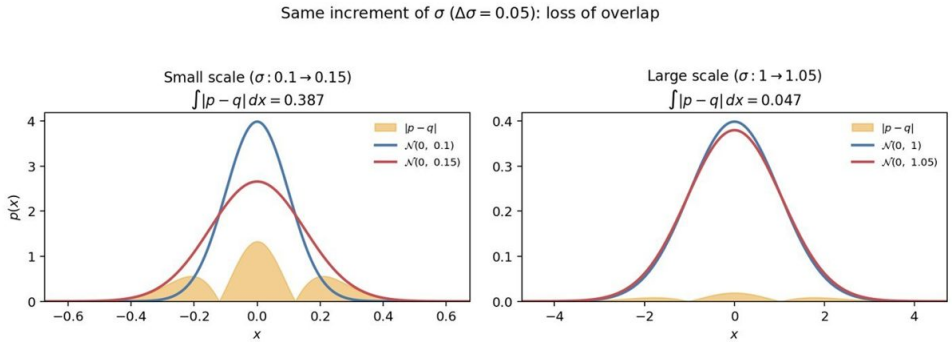


FIGURE 13. The same increase in standard deviation $\Delta\sigma = 0{,}05$. On the left, from 0,1 to 0,15, the profile changes considerably; on the right, from 1 to 1,05, it barely changes. Once again, the cost depends on the current scale of σ.

Through the identification with the Poincaré half-plane, the geodesics are known explicitly: they are vertical rays (pure changes of σ at fixed μ) or semicircles orthogonal to the axis $\sigma = 0$, after the change of coordinates $\xi = \mu/\sqrt{2}$. Figure 14 shows one such geodesic between $\mathcal{N}(-2, 1)$ and $\mathcal{N}(2, 1)$. To connect two means with the same standard deviation, the shortest path is not the horizontal translation: one first increases σ, translates μ where the metric is more accommodating, and then decreases it again. On the right, the densities along the curve make this movement visible.

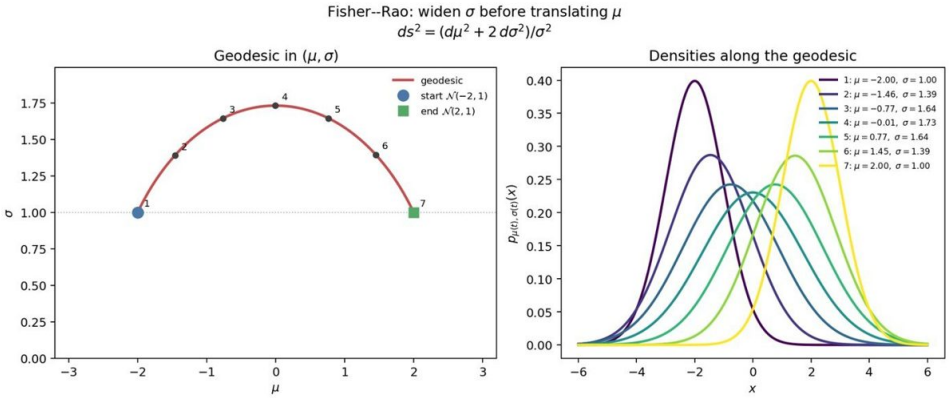


FIGURE 14. Fisher–Rao geodesic from $\mathcal{N}(-2,1)$ to $\mathcal{N}(2,1)$. On the left, the arc in (μ,σ); on the right, the densities at the numbered points. We increase σ before translating μ.

4.2. **Multivariate Gaussian**

In the multivariate case, the parametrization $p\colon \mathbb{R}^n \times S^n_{++}(\mathbb{R}) \to \mathcal{P}_+(\mathbb{R}^n)$ is given by

$$p_{\mu,\Sigma}(x) = \frac{1}{(2\pi)^{n/2}\,\det(\Sigma)^{1/2}}\, e^{-\frac{1}{2}(x-\mu)^\top \Sigma^{-1}(x-\mu)}.$$

Figure 15 shows four members of this family in $\mathbb{R}^2$, all centred at $\mu = 0$, with covariance matrices of different elongations and orientations. The graph of the density is drawn as a translucent surface; on the base plane, the level sets trace the ellipse associated with Σ.

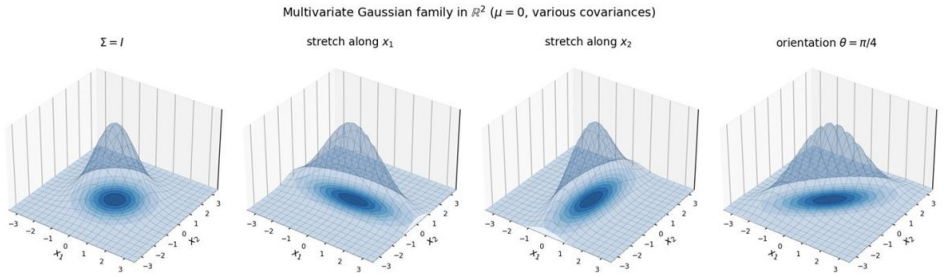


FIGURE 15. Four multivariate Gaussians in $\mathbb{R}^2$, centred at $\mu = 0$. From left to right: $\Sigma = I$, elongation along x_1, elongation along x_2, then orientation $\theta = \pi/4$. The surface represents the graph of $p_{0,\Sigma}$; the curves on the base plane are the level sets of the density.

To pull the Fisher–Rao metric back to the product $\mathbb{R}^n \times S^n_{++}$, we use the Fréchet differential of p.

4.2.1. **Differential**

Three identities from differential calculus are sufficient:

- $d(\det)(M)(H) = \det(M)\ \mathrm{tr}(M^{-1}H)$ for invertible M;
- $d(M \mapsto M^{-1})(M)(H) = -M^{-1}HM^{-1}$;
- since Σ^{-1} is symmetric, $d(x \mapsto x^\top \Sigma^{-1} x)(x)(y) = 2\, x^\top \Sigma^{-1} y$.

Here $d\mu$ and $d\Sigma$ denote tangent directions. Since $\mathbb{R}^n$ is a vector space, $T_\mu \mathbb{R}^n$ is canonically identified with $\mathbb{R}^n$. Likewise, S^n_{++} is an open subset of the vector space S^n of symmetric matrices, so $T_\Sigma S^n_{++}$ is canonically identified with S^n. Thus, $d\Sigma$ is symmetric, but need not itself be positive definite.

The differential with respect to μ is

$$d_\mu p_{\mu,\Sigma}(d\mu)(x) = \big((x-\mu)^\top \Sigma^{-1} d\mu\big)\, p_{\mu,\Sigma}(x), \qquad d\mu \in T_\mu \mathbb{R}^n = \mathbb{R}^n,$$

whereas the differential with respect to Σ, for $d\Sigma \in T_\Sigma S^n_{++} = S^n$, is

$$d_\Sigma p_{\mu,\Sigma}(d\Sigma)(x) = \frac{1}{2}\Big(-\mathrm{tr}(\Sigma^{-1} d\Sigma) + (x-\mu)^\top \Sigma^{-1} d\Sigma\, \Sigma^{-1}(x-\mu)\Big) p_{\mu,\Sigma}(x).$$

The total differential on the Cartesian product is the sum of the two:

$$\begin{aligned} dp_{\mu,\Sigma}(d\mu, d\Sigma)(x) = p_{\mu,\Sigma}(x)\Big[&(x-\mu)^\top \Sigma^{-1} d\mu \\ &+ \frac{1}{2}\big(-\mathrm{tr}(\Sigma^{-1} d\Sigma) + (x-\mu)^\top \Sigma^{-1} d\Sigma\, \Sigma^{-1}(x-\mu)\big)\Big]. \end{aligned}$$

4.2.2. **Metric**

The pullback metric is given by

$$\begin{aligned} &g_{\mu,\Sigma}\big((d\mu_1, d\Sigma_1), (d\mu_2, d\Sigma_2)\big) \\ &\qquad = \int_{\mathbb{R}^n} \frac{dp_{\mu,\Sigma}(d\mu_1, d\Sigma_1)(x)\ dp_{\mu,\Sigma}(d\mu_2, d\Sigma_2)(x)}{p_{\mu,\Sigma}(x)}\, dx. \end{aligned}$$

Let $A = \Sigma^{-1}$, $Y = X - \mu$, and

$$h(d\mu, d\Sigma)(y) = y^\top A\, d\mu + \frac{1}{2}\left[-\mathrm{tr}(A\, d\Sigma) + y^\top A\, d\Sigma\, A\, y\right].$$

The integral then becomes an expectation under $Y \sim \mathcal{N}(0, \Sigma)$:

$$g_{\mu,\Sigma}\big((d\mu_1, d\Sigma_1), (d\mu_2, d\Sigma_2)\big) = \mathbb{E}[h(d\mu_1, d\Sigma_1)(Y)\, h(d\mu_2, d\Sigma_2)(Y)]\,.$$

We now compute the metric block by block.

Mean block $g^{\mu\mu}$. It retains only the terms that are linear in Y:

$$\mathbb{E}\Big[(Y^\top A\, d\mu_1)(Y^\top A\, d\mu_2)\Big]\,.$$

Since $\mathbb{E}[YY^\top] = \Sigma$ by the definition of Y, this is equal to

$$d\mu_1^\top A\, \mathbb{E}[YY^\top]\, A\, d\mu_2 = d\mu_1^\top A \Sigma A\, d\mu_2 = d\mu_1^\top \Sigma^{-1} d\mu_2,$$

Thus, $g^{\mu\mu}(d\mu_1, d\mu_2) = d\mu_1^\top \Sigma^{-1} d\mu_2$.

Cross block $g^{\mu\Sigma}$. It involves a centred moment of order three:

$$\frac{1}{2}\,\mathbb{E}\Big[(Y^\top A\, d\mu)\,\Big(-\mathrm{tr}(A\, d\Sigma) + Y^\top A\, d\Sigma\, A\, Y\Big)\Big]\,.$$

The first term contains a centred moment of order one and the second a centred moment of order three. Both vanish because all odd moments of a centred Gaussian are zero. Hence $g^{\mu\Sigma}(d\mu, d\Sigma) = 0$, and the mean and covariance are orthogonal with respect to the Fisher–Rao metric.

Covariance block $g^{\Sigma\Sigma}$. Set $B_i = A\, d\Sigma_i\, A$. The Gaussian fourth-moment identity

$$\mathbb{E}\Big[(Y^\top B_1 Y)(Y^\top B_2 Y)\Big] = 2\ \mathrm{tr}(B_1\Sigma B_2\Sigma) + \mathrm{tr}(B_1\Sigma)\ \mathrm{tr}(B_2\Sigma).$$

Since $\mathbb{E}[Y^\top B_i Y] = \mathrm{tr}(B_i\Sigma) = \mathrm{tr}(A\, d\Sigma_i)$, the products of traces cancel when the product of the two scores is expanded. After multiplying by the factor $1/4$ from the two scores, this gives

$$g^{\Sigma\Sigma}(d\Sigma_1, d\Sigma_2) = \frac{1}{2}\ \mathrm{tr}\big(\Sigma^{-1} d\Sigma_1\, \Sigma^{-1} d\Sigma_2\big).$$

A detailed treatment of this Fisher metric and of the resulting Riemannian geometry of the multivariate normal model is given by Skovgaard [12].
Summary. The metric is block-diagonal:

$$g_{\mu,\Sigma} = \begin{pmatrix} g^{\mu\mu} & 0 \\ 0 & g^{\Sigma\Sigma} \end{pmatrix},$$

which can also be written in differential form as

$$ds^2 = d\mu^\top \Sigma^{-1} d\mu + \frac{1}{2}\ \mathrm{tr}\big(\Sigma^{-1} d\Sigma\, \Sigma^{-1} d\Sigma\big).$$

4.2.3. **Analysis of the metric**

In higher dimensions, interpreting the metric requires a little more care. Its mechanisms can nevertheless be disentangled using the spectral decomposition of Σ. Since Σ is symmetric positive definite, the spectral theorem gives

$$\Sigma = R\, D\, R^\top, \qquad R \in \mathrm{SO}(n), \quad D = \mathrm{diag}(\sigma_1^2, \dots, \sigma_n^2),$$

with $\sigma_i > 0$: σ_i is the standard deviation along the i-th column of R, that is, along the i-th principal axis of the Gaussian.

The spectral representation should be understood locally when it is used as a system of coordinates. When the eigenvalues are distinct, the principal axes can be chosen smoothly locally, up to the usual discrete ambiguities. When two eigenvalues coincide, however, the corresponding orientation is no longer determined by Σ: rotations inside the associated eigenspace leave the covariance matrix unchanged. Thus the variables (σ_i, R) are used below primarily to interpret the different components of the metric, rather than as global coordinates on S^n_{++}.

Three elementary motions then emerge: a shift of the mean at fixed covariance; a change in standard deviation along one or more directions, with the orientation and mean held fixed; and finally a rotation of these axes, with the standard deviations and mean held fixed. We illustrate these three motions using bivariate Gaussians, where the dimension remains low enough for the eye to follow. This parametrization by eigenvalues, orientation, and mean is the one used by Costa, Santos, and Strapasson [14] in the bivariate case.

To make this interpretation explicit, we express the metric g in terms of $dD \in T_D\mathrm{Diag}_{++}(n) = \mathrm{Diag}(n)$ and $dR \in T_R\mathrm{SO}(n) = \{R\Omega : \Omega^\top = -\Omega\}$. We then have

$$d\Sigma = R\, dD\, R^\top + dR\, D\, R^\top + R\, D\, dR^\top.$$

Expressing the variation of the mean in the basis that diagonalizes the covariance, $\eta = R^\top d\mu$, gives

$$\boxed{ds^2 = \sum_{i=1}^{n} \frac{\eta_i^2}{\sigma_i^2} + 2\sum_{i=1}^{n} \frac{d\sigma_i^2}{\sigma_i^2} + \sum_{1\le i<j\le n} \frac{(\sigma_i^2-\sigma_j^2)^2}{\sigma_i^2\sigma_j^2}\,\Omega_{ij}^2}$$

with $\Omega = R^\top dR \in \mathfrak{so}(n)$.

The metric is easier to interpret in this form. The first two terms correspond to the metric of the univariate Gaussian, summed over each of the coordinates associated with the eigenvectors that form the new basis. The third term, which couples the different standard deviations, corresponds to a change in the orientation of the Gaussian. Let us examine this in more detail.

4.2.4. **Change in the mean**

If we assume $d\Sigma = 0$, the metric reduces to

$$g_{\mu,\Sigma}(d\mu, d\mu) = d\mu^\top \Sigma^{-1} d\mu.$$

This is the squared Mahalanobis norm of $d\mu$ associated with Σ (figure 16). Some care is nevertheless required: this quadratic form describes an infinitesimal displacement at fixed Σ, not a distance between two Gaussians, and it does not imply that geodesics of the full model keep Σ constant. As in the univariate case, the shortest path between two Gaussians with the same covariance but different means involves spreading the covariance along the way (figures 17–19).

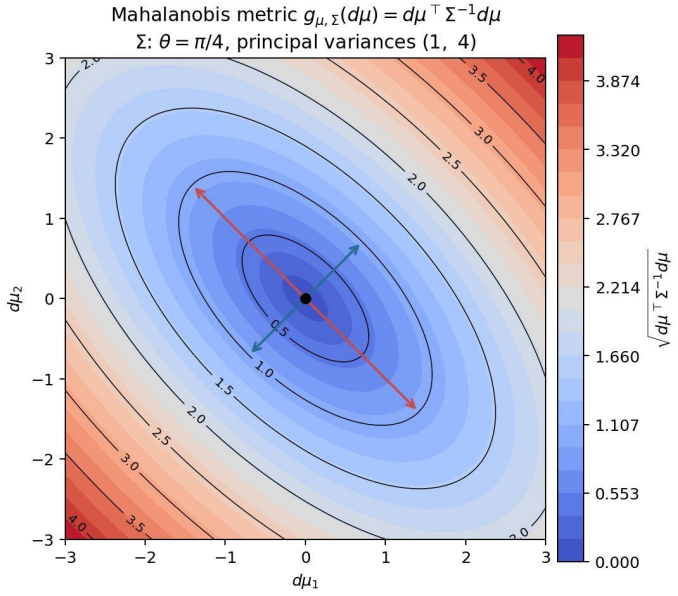


FIGURE 16. Metric $g_{\mu,\Sigma}$ on mean displacements, at fixed Σ. Here Σ has angle $\pi/4$ and principal variances $(1,\ 4)$. The level sets are Mahalanobis ellipses; the arrows mark the principal axes.

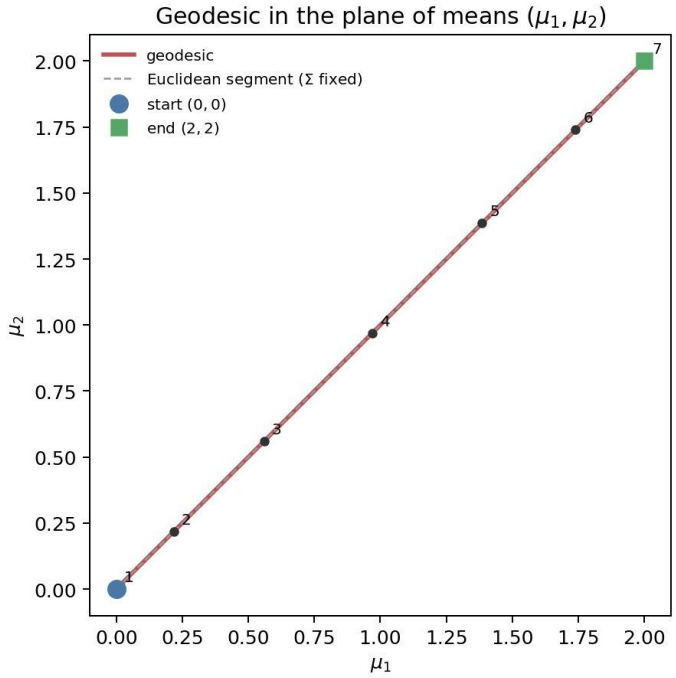


FIGURE 17. Numerical approximation of a Fisher–Rao geodesic from $\mathcal{N}((0,0),\Sigma)$ to $\mathcal{N}((2,2),\Sigma)$, projected onto the (μ_1,μ_2) plane, with θ fixed at $\pi/4$. The mean displacement follows the line segment; it is Σ that changes.

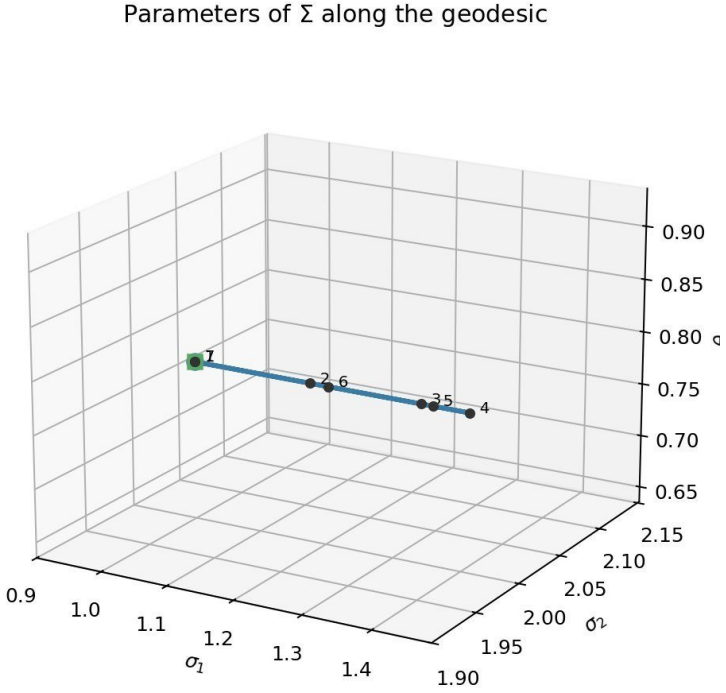


FIGURE 18. Parameters $(\sigma_1, \sigma_2, \theta)$ of Σ along the same curve. The standard deviations move away from their endpoint values in the middle of the path (spreading), then return to them; θ is constrained to $\pi/4$.

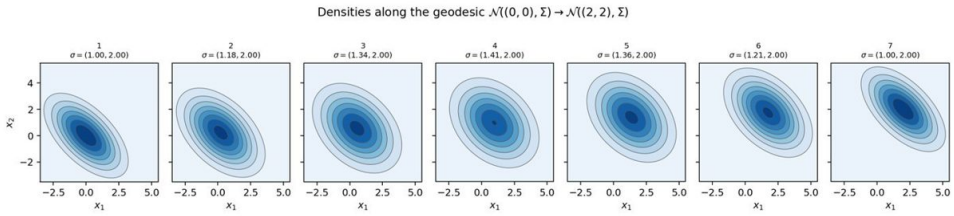


FIGURE 19. Densities along the same curve. From 1 to 7, the mean shifts while the Gaussian widens and then narrows again, as in the univariate case.

4.2.5. **Change in standard deviation**

If only the standard deviations are allowed to vary ($d\mu = 0$, $dR = 0$), the metric reduces to

$$ds^2 = 2\sum_{i=1}^{n} \frac{d\sigma_i^2}{\sigma_i^2}.$$

Each eigendirection therefore carries its own copy of the univariate σ block; in the coordinates $(\log \sigma_i)$, this is a diagonal Euclidean metric.

Consider the path between two centred Gaussians with the same orientation $\theta = \pi/4$, whose principal variances change from $(1,\ 4)$ to $(4,\ 1)$, that is, with (σ_1, σ_2) moving from $(1,\ 2)$ to $(2,\ 1)$. This time, μ and θ remain fixed along the geodesic; only the pair (σ_1, σ_2) changes. In the plane of standard deviations, the path is not the Euclidean line segment: it is the arc $\sigma_1(t) = 2^t$, $\sigma_2(t) = 2^{1-t}$ (figure 20), and the densities gradually exchange their elongations (figure 21).

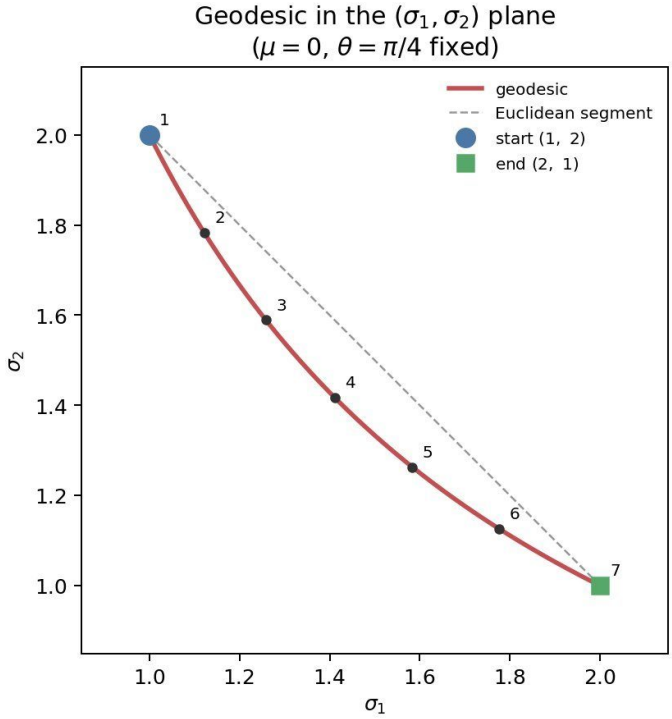


FIGURE 20. Fisher–Rao geodesic in the (σ_1, σ_2) plane, with $\mu = 0$ and $\theta = \pi/4$ fixed. The path is a hyperbola (a straight line in logarithmic coordinates), distinct from the Euclidean line segment.

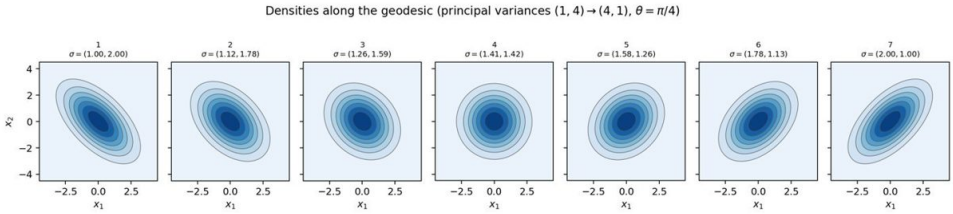


FIGURE 21. Densities along the geodesic. The mean and orientation remain fixed; the principal variances exchange their values, changing from $(1,\ 4)$ to $(4,\ 1)$.

4.2.6. **Change in orientation**

Finally, if only the orientation is allowed to vary ($d\mu = 0$, $d\sigma_i = 0$), only the third term remains:

$$ds^2 = \sum_{1 \le i < j \le n} \frac{(\sigma_i^2 - \sigma_j^2)^2}{\sigma_i^2 \sigma_j^2} \, \Omega_{ij}^2.$$

In dimension two, Ω reduces to a single degree of freedom, the angle θ, and the coefficient multiplying $d\theta^2$ depends only on the standard deviations. As long as they remain fixed, $g_{\theta\theta}$ is therefore constant: the orientation geodesic has uniform angular speed.

We start from a centred Gaussian with elongation $(1,\ 4)$, oriented at $\pi/4$, and take it to $3\pi/4$ without changing either μ or the σ_i. The densities rotate in place (figure 22); the graph of $\theta(t)$ is a straight line, and the Riemannian speed $\sqrt{g_{\theta\theta}}\,|\dot\theta|$ remains constant (figure 23).

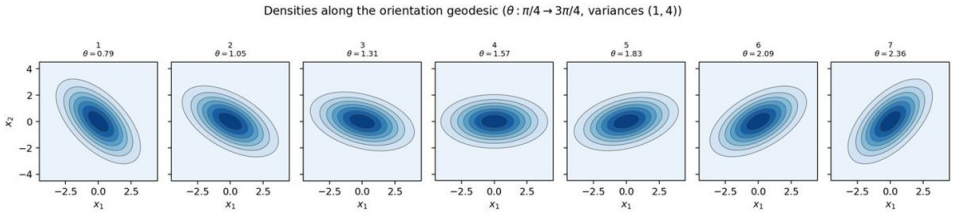


FIGURE 22. Densities along the orientation geodesic. The mean and standard deviations are fixed; only θ changes from $\pi/4$ to $3\pi/4$.

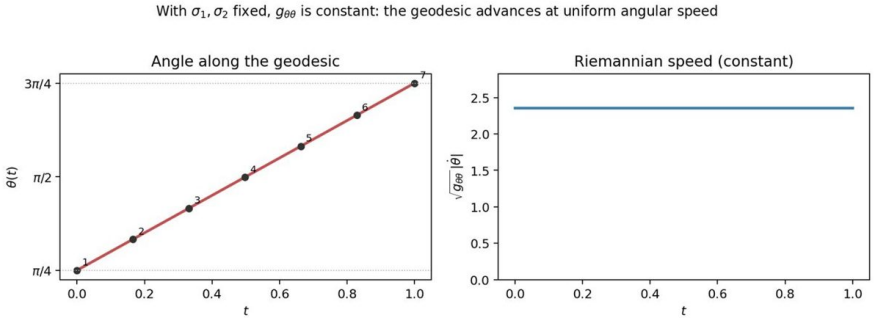


FIGURE 23. On the left, $\theta(t)$ along the geodesic (affine); on the right, the Riemannian speed, which is constant because $g_{\theta\theta}$ does not depend on θ.

4.2.7. **When everything moves together**

The three previous motions combine as soon as two arbitrary Gaussians are connected. The mean, standard deviations, and orientation then move together, each at the rate dictated by its part of the metric. Eriksen [15] integrates geodesics with a prescribed initial velocity using the exponential

of a $(2n+1) \times (2n+1)$ matrix; Calvo and Oller [16] give an explicit form. Connecting two Gaussians is a boundary-value problem: once $n \geq 2$, there is no elementary formula for either the path or the distance. Here we approximate this boundary-value problem by minimizing a discrete Riemannian energy (section 9). As an illustration, we connect

$$\mathcal{N}\big((0,0),\Sigma_1\big) \quad \text{to} \quad \mathcal{N}\big((2,2),\Sigma_2\big),$$

where Σ_1 has elongation $(1,\ 4)$ and orientation $\pi/4$, while Σ_2 has elongation $(4,\ 1)$ and orientation $3\pi/4$. Figure 24 shows the densities along the geodesic: the mass shifts, stretches, and rotates at the same time. Halfway along the path, the two standard deviations draw closer; the Gaussian becomes almost isotropic and $g_{\theta\theta}$ collapses, allowing the orientation to rotate at little cost before closing onto the target ellipse. This change can be seen clearly in the space $(\sigma_1, \sigma_2, \theta)$ (figure 25): the path rises in θ mainly when $\sigma_1 \approx \sigma_2$.

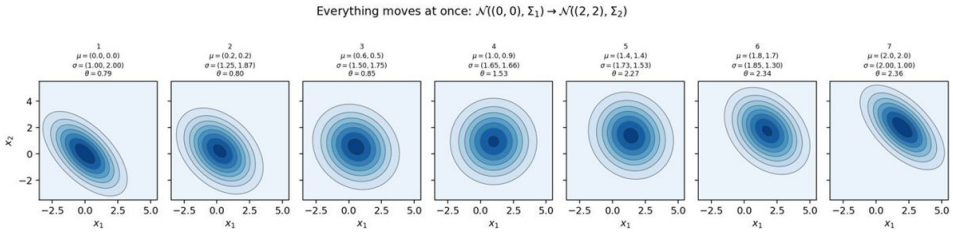


FIGURE 24. Numerical approximation of a Fisher–Rao geodesic between two general bivariate Gaussians (discrete energy minimization). At each step, μ, (σ_1, σ_2), and θ evolve simultaneously.

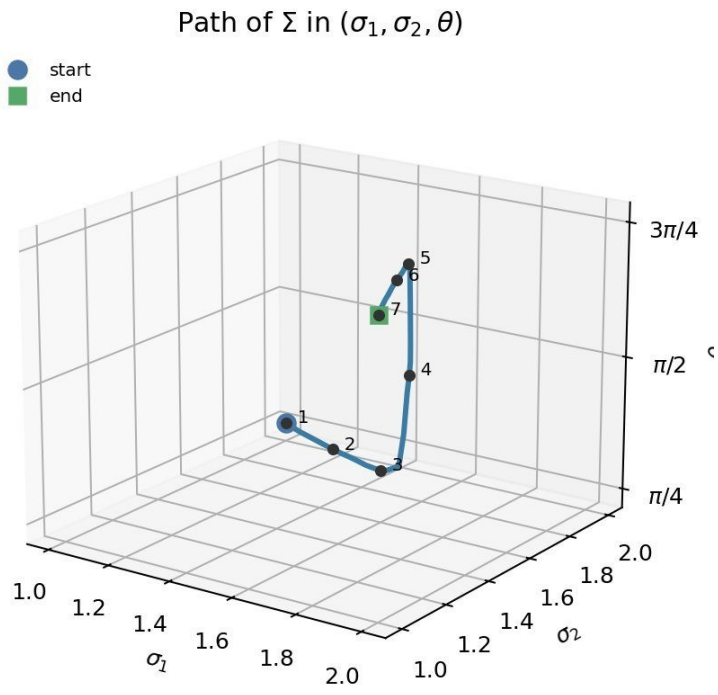


FIGURE 25. The same curve, projected into $(\sigma_1, \sigma_2, \theta)$. The increase in θ is concentrated near the diagonal $\sigma_1 = \sigma_2$, where rotation costs the least.

5. Elliptical Laplace family

We now consider the family

$$p_{\mu,\Sigma}(x) = c_n \, |\Sigma|^{-1/2} \exp\!\left(-\tfrac{1}{2}\sqrt{(x-\mu)^\top \Sigma^{-1}(x-\mu)}\right),$$

with $c_n = \Gamma(n/2)/(2^{n+1}\pi^{n/2}\Gamma(n))$, so that p is a density. Here $\Sigma \in S^n_{++}$ is a *dispersion* parameter: it is not the covariance. A radial calculation gives $\mathrm{Cov}(X) = 4(n+1)\,\Sigma$. We follow the same interpretation as for the Gaussian, moving more quickly whenever the mechanisms are the same.

5.1. Univariate Laplace

For $n = 1$, $\Sigma = \sigma^2$ and $c_1 = 1/4$, hence

$$p_{\mu,\sigma}(x) = \frac{1}{4\sigma} \exp\!\left(-\frac{|x-\mu|}{2\sigma}\right).$$

Figure 26 shows three centred members with different dispersions: a peak at μ and exponential tails.

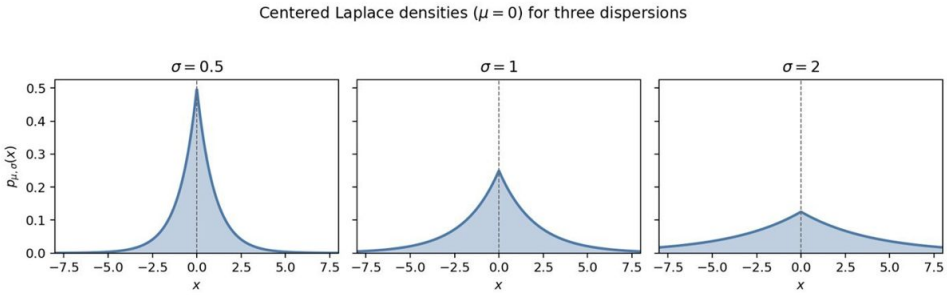


FIGURE 26. Three centred univariate Laplace densities ($\mu = 0$), for $\sigma \in \{0{,}5,\ 1,\ 2\}$.

5.1.1. **Differential**

The parameters are again a location μ and a dispersion $\sigma > 0$. Away from $x = \mu$, where $|\cdot|$ is not differentiable, the chain rule gives

$$\partial_\mu p_{\mu,\sigma}(x) = \frac{\operatorname{sign}(x-\mu)}{2\sigma}\, p_{\mu,\sigma}(x), \qquad \partial_\sigma p_{\mu,\sigma}(x) = \left(-\frac{1}{\sigma} + \frac{|x-\mu|}{2\sigma^2}\right) p_{\mu,\sigma}(x),$$

or equivalently, for the scores,

$$\partial_\mu \log p = \frac{\operatorname{sign}(x-\mu)}{2\sigma}, \qquad \partial_\sigma \log p = -\frac{1}{\sigma} + \frac{|x-\mu|}{2\sigma^2}.$$

5.1.2. **Metric**

With $|X - \mu| \sim \mathrm{Exp}(1/(2\sigma))$, we obtain $g_{\mu\sigma} = 0$ and

$$g_{\mu\mu} = \frac{1}{4\sigma^2}, \qquad g_{\sigma\sigma} = \frac{1}{\sigma^2}, \qquad \text{that is} \qquad ds^2 = \frac{d\mu^2}{4\sigma^2} + \frac{d\sigma^2}{\sigma^2}.$$

The change $(\xi, \eta) = (\mu/2,\ \sigma)$ gives $ds^2 = (d\xi^2 + d\eta^2)/\eta^2$: this is again the Poincaré hyperbolic half-plane, as for the Gaussian, but without the factor 2 in front of $d\sigma^2$ and with a rescaling of μ. The geometry is the same, but the coefficients differ.

5.1.3. **Analysis of the metric**

The interpretation is the same as in section 3.2: the cost of a displacement depends on the current dispersion. Moving from $L(0,\ 0{,}1)$ to $L(0{,}05,\ 0{,}1)$ misaligns the densities much more than the same $\Delta\mu$ with $\sigma = 1$ (figure 27); the same holds for a fixed $\Delta\sigma$ (figure 28).

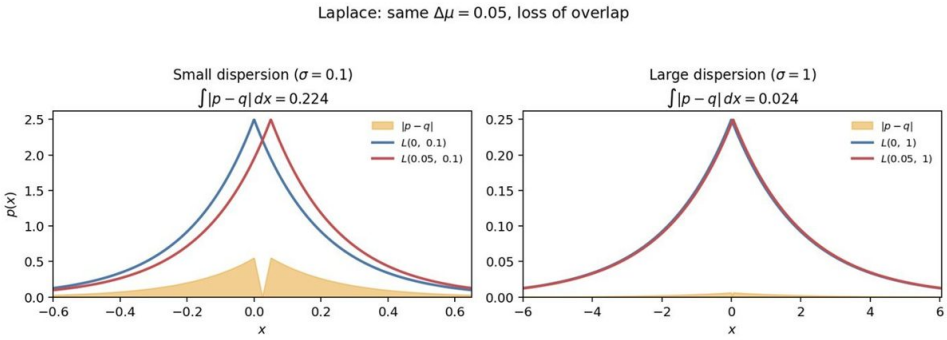


FIGURE 27. Univariate Laplace: the same mean shift $\Delta\mu = 0{,}05$. On the left, $\sigma = 0{,}1$; on the right, $\sigma = 1$. The orange area marks $|p - q|$.

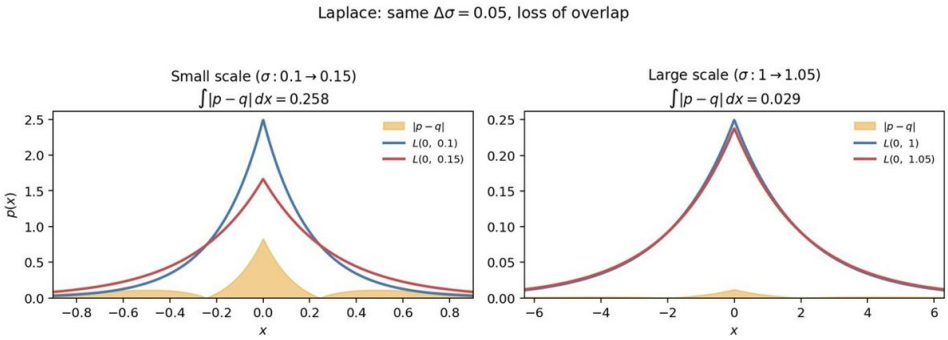


FIGURE 28. Univariate Laplace: the same increase $\Delta\sigma = 0{,}05$. The misalignment is large at small scale and small at large scale.

5.2. **Multivariate Laplace**

In dimension two, figure 29 shows four members of the family, all centred at $\mu = 0$: isotropic dispersion, axis-aligned elongation, then orientations $\pi/4$ and $3\pi/4$. The level sets are the same ellipses as in the Gaussian case, since the density depends on x only through $(x - \mu)^\top \Sigma^{-1}(x - \mu)$. What changes is the decay profile along a radius: $e^{-r/2}$ here, compared with $e^{-r^2/2}$ for the Gaussian, hence the more pronounced peak at μ and the heavier tails.

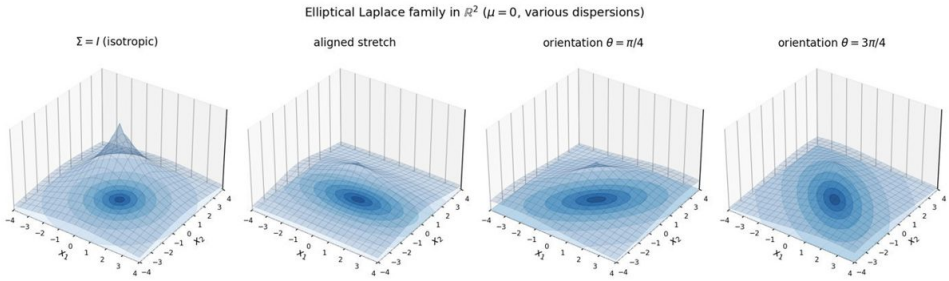


FIGURE 29. Four elliptical Laplace densities in $\mathbb{R}^2$, centred at $\mu = 0$. From left to right: $\Sigma = I$, axis-aligned elongation, orientation $\pi/4$, orientation $3\pi/4$.

5.2.1. **Differential**

Let $Y = X - \mu$, $A = \Sigma^{-1}$, and $r = \sqrt{Y^\top A Y}$. Away from $Y = 0$, the chain rule gives

$$d_\mu \log p(d\mu) = \frac{Y^\top A\, d\mu}{2r},$$
$$d_\Sigma \log p(d\Sigma) = -\tfrac{1}{2}\operatorname{tr}(A\, d\Sigma) + \frac{Y^\top A\, d\Sigma\, A\, Y}{4r},$$

and therefore $dp = p\, d\log p$, as for the Gaussian.

5.2.2. **Metric**

Use the radial decomposition $Y = \Sigma^{1/2}(rU)$, where U is uniform on the sphere and r has density proportional to $e^{-r/2}r^{n-1}$. The radial variable is therefore Gamma distributed, and the moments needed below are

$$\mathbb{E}[r] = 2n, \qquad \mathbb{E}[r^2] = 4n(n+1), \qquad \mathbb{E}[UU^\top] = \frac{1}{n} I.$$

For two symmetric matrices C_1 and C_2, the fourth spherical moment is

$$\mathbb{E}\Big[(U^\top C_1 U)(U^\top C_2 U)\Big] = \frac{2\operatorname{tr}(C_1 C_2) + \operatorname{tr}(C_1)\operatorname{tr}(C_2)}{n(n+2)}.$$

For the mean block, the radial variable cancels from the score:

$$d_\mu \log p(d\mu) = \frac{1}{2}\, U^\top \Sigma^{-1/2} d\mu.$$

It follows from $\mathbb{E}[UU^\top] = I/n$ that

$$g^{\mu\mu}(d\mu_1, d\mu_2) = \frac{1}{4n}\, d\mu_1^\top \Sigma^{-1} d\mu_2.$$

The product of this linear score with the covariance score is odd in U, so the cross block $g_{\mu\Sigma}$ vanishes.

For the covariance block, set $C_i = \Sigma^{-1/2} d\Sigma_i \Sigma^{-1/2}$ and $t_i = \operatorname{tr}(C_i)$. The corresponding score is

$$s_i = -\frac{1}{2} t_i + \frac{r}{4}\, U^\top C_i U.$$

Using the radial and spherical moments above, its product satisfies

$$\begin{aligned}\mathbb{E}[s_1 s_2] &= \frac{1}{4} t_1 t_2 - \frac{1}{2} t_1 t_2 + \frac{n+1}{4(n+2)} \left(2 \operatorname{tr}(C_1 C_2) + t_1 t_2\right) \\ &= \frac{n+1}{2(n+2)} \operatorname{tr}(C_1 C_2) - \frac{1}{4(n+2)} t_1 t_2.\end{aligned}$$

Thus,

$$g^{\Sigma\Sigma}(d\Sigma_1, d\Sigma_2) = \frac{n+1}{2(n+2)} \operatorname{tr}(\Sigma^{-1} d\Sigma_1 \, \Sigma^{-1} d\Sigma_2) - \frac{1}{4(n+2)} \operatorname{tr}(\Sigma^{-1} d\Sigma_1) \operatorname{tr}(\Sigma^{-1} d\Sigma_2).$$

The information matrix of general multivariate elliptical families is treated in detail by Mitchell [13].

Altogether,

$$ds^2 = a_n \, d\mu^\top \Sigma^{-1} d\mu + b_n \operatorname{tr}(\Sigma^{-1} d\Sigma \, \Sigma^{-1} d\Sigma) + c_n \big[\operatorname{tr}(\Sigma^{-1} d\Sigma)\big]^2,$$

with

$$a_n = \frac{1}{4n}, \qquad b_n = \frac{n+1}{2(n+2)}, \qquad c_n = -\frac{1}{4(n+2)}.$$

For the Gaussian, we had $a_n = 1$, $b_n = 1/2$, $c_n = 0$. The c_n term is new: it depends only on the trace, that is, on the isotropic scaling mode.

5.3. Analysis of the metric

As for the Gaussian, $\Sigma = R\,D\,R^\top$ with $D = \operatorname{diag}(\sigma_1^2, \dots, \sigma_n^2)$, $\eta = R^\top d\mu$, and $\Omega = R^\top dR \in \mathfrak{so}(n)$. Substituting into ds^2 gives

$$\boxed{\begin{aligned} ds^2 = a_n \sum_{i=1}^n \frac{\eta_i^2}{\sigma_i^2} + \frac{2(n+1)}{n+2} \sum_{i=1}^n \frac{d\sigma_i^2}{\sigma_i^2} - \frac{1}{n+2} \Big(\sum_{i=1}^n \frac{d\sigma_i}{\sigma_i} \Big)^2 \\ + \frac{n+1}{n+2} \sum_{1 \le i < j \le n} \frac{(\sigma_i^2 - \sigma_j^2)^2}{\sigma_i^2 \sigma_j^2} \, \Omega_{ij}^2. \end{aligned}}$$

As we saw for the multivariate Gaussian, a deformation of the density can be understood through three motions: a change in the mean, a change in spread (the σ_i), and a change in orientation.

5.3.1. Mean and orientation

At fixed spread, the metric is of the same type as for the Gaussian. For the mean, we recover a Mahalanobis term $a_n \, d\mu^\top \Sigma^{-1} d\mu$; only the constant a_n changes. The same caution applies: geodesics of the full model have no reason to keep Σ constant, and the relative weight of a_n compared with the Σ terms is no longer the same as for the Gaussian. For the orientation, the rotational cost has the same form, up to the prefactor $(n+1)/(n+2)$: it vanishes whenever $\sigma_i = \sigma_j$ and grows with the anisotropy $|\sigma_i^2 - \sigma_j^2|$. Relative to the anisotropy weight $\frac{2(n+1)}{n+2}$, this prefactor is the same as in the Gaussian case.

5.3.2. **Dispersion**

The term governing changes in dispersion, however, is no longer a multiple of $\sum_i (d\sigma_i/\sigma_i)^2$. Denoting the isotropic component by $\overline{d\sigma/\sigma} = \frac{1}{n}\sum_i d\sigma_i/\sigma_i$,

$$ds^2_{\text{disp}} = n\big(\overline{d\sigma/\sigma}\big)^2 + \frac{2(n+1)}{n+2}\sum_{i=1}^{n}\Big(\frac{d\sigma_i}{\sigma_i} - \overline{d\sigma/\sigma}\Big)^2.$$

Both coefficients are positive: $c_n < 0$ does not prevent the block from being positive definite. The first term is the change in global scale (the geometric mean of the σ_i); the second is the change in anisotropy, orthogonal to the isotropic mode. In these coordinates, $c_n\big[\operatorname{tr}(\Sigma^{-1}d\Sigma)\big]^2$ is equal to $-\frac{1}{n+2}\big(\sum_i d\sigma_i/\sigma_i\big)^2$ and acts only on the first mode, whose weight it lowers. For an exchange between axes, $\sum_i d\sigma_i/\sigma_i = 0$, so this term vanishes: it does not penalize the exchange. For the Gaussian, $c_n = 0$, and we recover $2\sum_i (d\sigma_i/\sigma_i)^2$, with the same weight for both modes. Here the global scale is relatively less costly than the change in shape.

In summary, the mean and orientation have the same form as in the Gaussian case, up to prefactors. The new term is c_n, and it couples the σ_i only through their sum. Geodesics of the full model also depend on the relative weights of a_n, the isotropic mode, the anisotropy, and the rotation, not only on this coupling.

To make this difference visible, we use the endpoints from the Gaussian example in which everything moves:

$$(\mu_0, \sigma_{1,0}, \sigma_{2,0}, \theta_0) = \big((0,0), 1, 2, \pi/4\big), \qquad (\mu_1, \sigma_{1,1}, \sigma_{2,1}, \theta_1) = \big((2,2), 2, 1, 3\pi/4\big).$$

Figure 30 superimposes the two geodesics in $(\sigma_1, \sigma_2, \theta)$. Their endpoints are identical, and the Gaussian curve is exactly the one from the previous section. The coupling specific to the Laplace dispersion parameters modifies the path followed before and during the rotation.

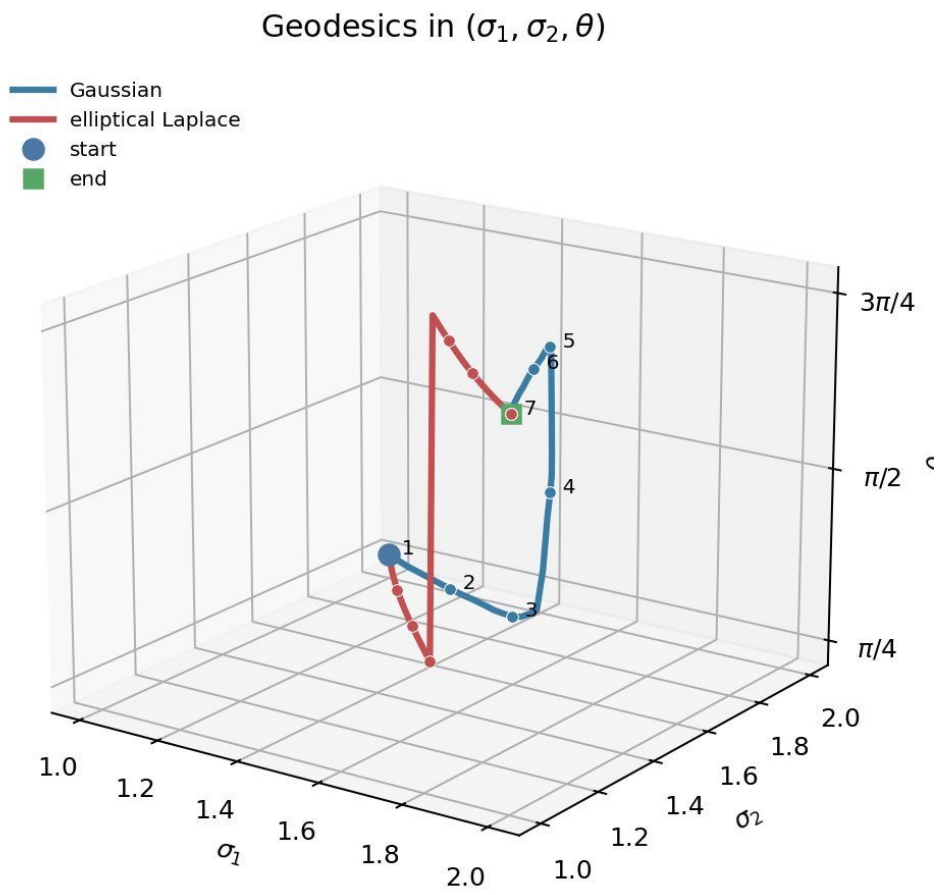


FIGURE 30. Numerical approximations of the Fisher–Rao geodesics between the same endpoint parameters, projected into $(\sigma_1, \sigma_2, \theta)$. Gaussian in blue, elliptical Laplace in red.

Figure 31 shows the corresponding densities at the same seven stages as figure 24: Gaussian on the top, elliptical Laplace on the bottom. The two families adopt the same basic strategy, already visible in the metric: rotation is paid for only near isotropy, where the coefficient $(\sigma_1^2-\sigma_2^2)^2/(\sigma_1^2\sigma_2^2)$ vanishes, and the displacement of the mean takes place during this passage. What distinguishes them is how they get there. For the Laplace family, the isotropic mode is less costly than the change in anisotropy, while rotation and anisotropy retain the same relative weight as in the Gaussian case; a_n is also smaller. The path in $(\sigma_1, \sigma_2, \theta)$ is no longer the same: the Laplace family comes closer to isotropy before rotating, then turns abruptly. The Gaussian begins to rotate earlier (stage 3) and spreads the rotation over two stages.

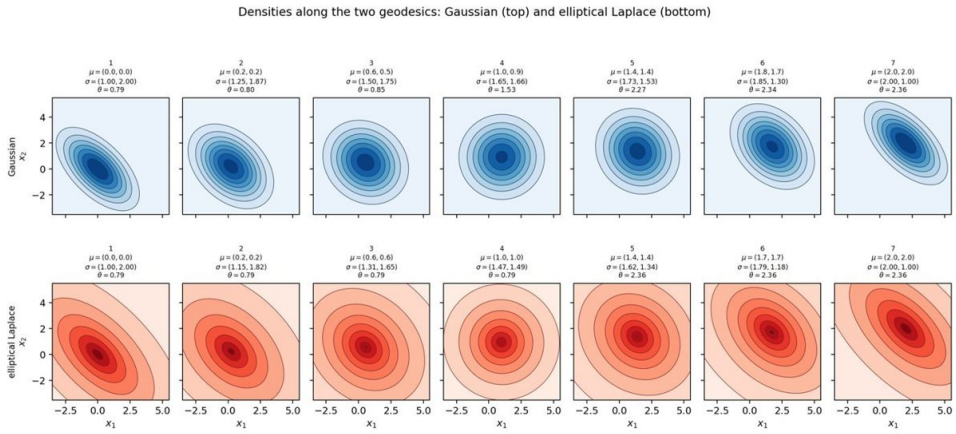


FIGURE 31. Evolution of the densities along the two numerical curves, at the same seven stages: Gaussian on the top, elliptical Laplace on the bottom. Each panel shows the values of μ, (σ_1, σ_2), and θ.

6. **Linear Gibbs family**

We consider densities of the form

$$p_\theta(x) = \frac{1}{Z_\theta}\, e^{-\phi(x)^\top \theta}, \qquad Z_\theta = \int_E e^{-\phi(x)^\top \theta}\, dx,$$

where θ belongs to an open set $\Theta \subset \mathbb{R}^k$ contained in the interior of the natural parameter domain

$$\{\theta \in \mathbb{R}^k : Z_\theta < \infty\},$$

and $\phi\colon E \to \mathbb{R}^k$ is measurable, so that $\phi(X)$ is a random variable (its law is the push-forward of p_θ by ϕ). We further assume that ϕ has finite moments of order up to three under p_θ, locally uniformly in θ. These assumptions justify the differentiations under the integral sign used below. This is the usual linear exponential family, written here in Gibbs form; see Amari–Nagaoka [17] and Nielsen [18].

In dimension one, we consider three choices of ϕ (figure 32): linear, with $\phi(x) = x$; polynomial, with $\phi(x) = (x, x^2)$; and $\phi(x) = (x^2, \cos x)$ on a short interval. The normalization constants are computed on a bounded portion of E.

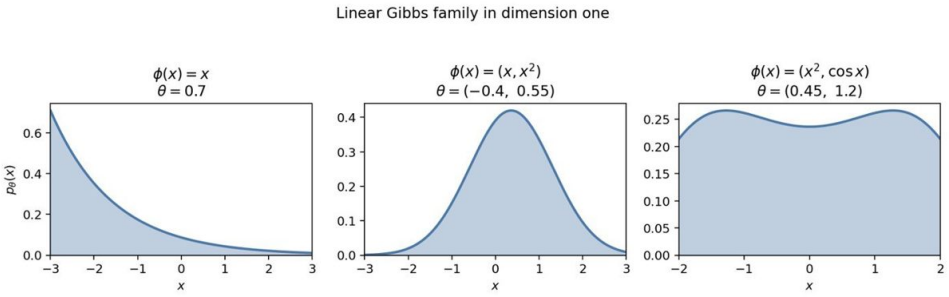


FIGURE 32. Three linear Gibbs families in dimension one, for fixed ϕ and θ. From left to right: $\phi(x) = x$, $\phi(x) = (x, x^2)$, $\phi(x) = (x^2, \cos x)$.

In dimension two (figure 33), we consider the linear case $\phi(x) = (x_1, x_2)$; a scalar quadratic statistic $\phi(x) = x_1^2 + x_1x_2 + x_2^2$; and $\phi(x) = (\cos(x_1^2 + x_2^2), \sin(4x_1x_2) + x_1 + x_2)$.

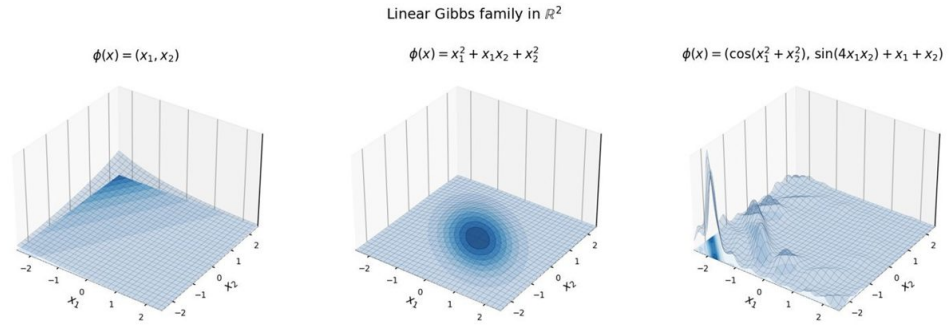


FIGURE 33. Three linear Gibbs densities in $\mathbb{R}^2$. From left to right: $\phi(x) = (x_1, x_2)$, $\phi(x) = x_1^2 + x_1x_2 + x_2^2$, $\phi(x) = (\cos(x_1^2 + x_2^2), \sin(4x_1x_2) + x_1 + x_2)$.

6.1. **Differential**

Let us differentiate p_θ with respect to θ. Since Z_θ depends on θ,

$$dp_\theta(d\theta)(x) = -p_\theta(x)\left(\phi(x)^\top d\theta + \frac{\nabla_\theta Z_\theta^\top d\theta}{Z_\theta}\right).$$

Now $\nabla_\theta Z_\theta = -Z_\theta\, \mathbb{E}_\theta[\phi(X)]$, so the second term is $-\mathbb{E}_\theta[\phi]^\top d\theta$ and

$$dp_\theta(d\theta) = -p_\theta\left(\phi - \mathbb{E}_\theta[\phi]\right)^\top d\theta.$$

6.2. **Metric**

The pullback metric follows:

$$\begin{aligned} g_\theta(d\theta_1, d\theta_2) &= \int_E \frac{dp_\theta(d\theta_1)\, dp_\theta(d\theta_2)}{p_\theta}\, dx \\ &= \mathbb{E}_\theta\Big[(\phi(X) - \mathbb{E}_\theta\phi)^\top d\theta_1\ (\phi(X) - \mathbb{E}_\theta\phi)^\top d\theta_2\Big] \\ &= d\theta_1^\top \operatorname{Cov}_\theta(\phi(X))\, d\theta_2. \end{aligned}$$

The Fisher–Rao geometry on the parameter space is therefore entirely encoded in $\operatorname{Cov}_\theta(\phi(X))$, the covariance of $\phi(X)$ when X has density p_θ.

By construction, $\operatorname{Cov}_\theta(\phi(X))$ is a symmetric positive-semidefinite matrix. It is positive definite when the components of ϕ are affinely independent, that is, when no non-trivial linear combination $v^\top\phi(x)$ is constant almost everywhere. In other words, the image of ϕ in $\mathbb{R}^k$ must not be contained in a proper affine subspace. The example $\phi(x) = (x, x^2)$ traces a parabola, and the condition is satisfied; with $\phi(x) = (x, 5x)$, the image is a line and the covariance is degenerate.

This shows the double role of ϕ: it determines the random variable whose covariance is taken, and it enters into p_θ, hence into the law under which this covariance is computed.

It is also the Hessian of $\log Z_\theta$:

$$\boxed{g_\theta = \operatorname{Hess}_\theta \log Z_\theta = \operatorname{Cov}_\theta(\phi(X))}$$

In general, a Riemannian metric amounts to choosing an inner product at every point: $k(k+1)/2$ functions of θ. Here everything comes from a single convex function, $\log Z_\theta$. We no longer choose g directly: we choose a potential, and g is its Hessian.

The total symmetry of $\partial_k g_{ij}$ follows immediately. These are the third derivatives of $\log Z_\theta$, and Schwarz's theorem makes them independent of the order of the indices. Conversely, in affine coordinates, $g = \operatorname{Hess}\psi$ if and only if

$$\partial_k g_{ij} = \partial_i g_{kj} = \partial_j g_{ik}$$

everywhere. This is the integrability condition, sometimes called the Codazzi condition: the way $g(e_i, e_j)$ changes when moving along e_k does not depend on which index plays which role. A general metric need not integrate to a potential.

This does not prevent the eigenaxes from rotating. For $\psi = \cosh\theta_1 + \cosh\theta_2 + 0{,}3\,\theta_1\theta_2$, the angle φ of the eigenvectors satisfies $\tan(2\varphi) = 0{,}6/(\cosh\theta_1 - \cosh\theta_2)$: the inclination varies with θ. What is ruled out is a field g for which the $\partial_k g_{ij}$ are not totally symmetric.

Figure 34 illustrates this in the (θ_1, θ_2) plane. The ellipses are circles of the metric: the vectors v such that $g_\theta(v, v) = r^2$, with r slightly smaller than 1 so that the field fits on the grid. On the left, $g = \operatorname{Hess}\psi$ for the convex potential above (an example, not $\log Z_\theta$). The ellipses stretch with ψ, while the background shows the level sets of ψ. On the right is a metric with constant eigenvalues, whose eigenaxes rotate only with θ_2. The background shows $\partial_2 g_{11} - \partial_1 g_{12}$, one component of the obstruction. It is non-zero: this field does not come from any potential.

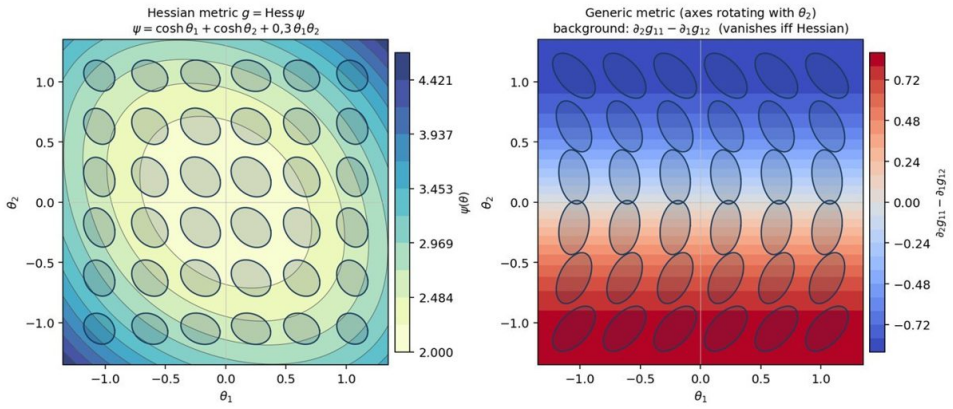


FIGURE 34. Circles of the metric in the (θ_1, θ_2) plane. On the left, $g = \mathrm{Hess}\,\psi$: the background shows the level sets of ψ. On the right, a field with fixed eigenvalues whose axes rotate with θ_2; the background, $\partial_2 g_{11} - \partial_1 g_{12}$, is the Schwarz obstruction, which vanishes if and only if g is a Hessian metric.

This rigidity has a counterpart. The gradient of the potential provides a second coordinate system, dual to the first:

$$\eta = -\nabla_\theta \log Z_\theta = \mathbb{E}_\theta[\phi].$$

In these coordinates, g^{-1} is the Hessian of the Legendre potential. We therefore have two affine charts, and the straight lines in each are distinguished paths of the dual Hessian geometry; they are not the Riemannian geodesics of g. This is the intuitive content of the dually flat geometry of exponential families [17, 18]. For Hessian geometry in general, see Shima [20]; for what this constraint rules out among Riemannian metrics, see Amari–Armstrong [21].

6.3. Analysis of the metric

The formula $g_\theta(d\theta, d\theta) = \mathrm{Var}_\theta\big(\phi(X)^\top d\theta\big)$ reads as follows. A displacement $d\theta$ is costly when the combination $\phi^\top d\theta$ fluctuates a lot under p_θ, and cheap when it is almost constant over the x that the law gives weight to. This is the reading of section 3.2, brought back onto Θ: $dp_\theta(d\theta)/p_\theta = -\big(\phi - \mathbb{E}_\theta\phi\big)^\top d\theta$, so the ratio u/p is none other than the centred energy gap. Fisher–Rao pays for these fluctuations, not the mean level, which Z_θ absorbs.

Concretely, ϕ chooses what θ is allowed to see. Two regions of E that ϕ does not distinguish remain on an equal footing, whatever the parameters. With $\phi(x) = x$, one can only tilt the density linearly; $\phi(x) = (x, x^2)$ adds control of the spread; $\phi(x) = \cos x$ sees only one oscillation, and a bounded one. Since ϕ also enters p_θ, the effective support moves with θ: increasing θ_i penalizes the x where ϕ_i is large, the mass concentrates elsewhere, and $\mathrm{Cov}_\theta(\phi)$ changes. The metric is not constant on Θ.

When p_θ spreads over a wide range of values of ϕ, each step of θ deforms the density a great deal. When it concentrates, typically for $|\theta|$ large, $\phi(X)$ freezes near a minimizer of $\phi^\top\theta$, the variance drops, and the same $d\theta$ in natural coordinates costs less and less. The case $\phi(x) = x$ on $\mathbb{R}_+$, the exponential law, makes this readable: $g_\theta = 1/\theta^2$, which blows up as $\theta \to 0$ (the density flattens out) and dies out as $\theta \to +\infty$ (all the mass is already glued to the origin).

On unbounded E, this blow-up often occurs at the boundary of the natural domain $\Theta = \{\theta : Z_\theta < \infty\}$. For $\phi(x) = x^2$ on $\mathbb{R}$, Z_θ is finite only if $\theta > 0$; as $\theta \downarrow 0$, the density spreads, and $\mathrm{Var}_\theta(X^2)$ goes to infinity. This is $\sigma \to +\infty$ in the Gaussian, read here in the natural parameter. The figures in this section, for their part, normalize on a bounded piece of E: Z_θ stays finite, the metric does not go to infinity. What remains are the anisotropies and the near-degeneracies.

If the image of ϕ is almost contained in an affine subspace, an eigenvalue of $\mathrm{Cov}_\theta(\phi)$ is tiny: the corresponding direction of θ adds almost a constant to the energy, and so hardly changes p_θ. This is the case of $\phi(x) = (x^2, \cos x)$ on a short interval, where $\cos x \approx 1 - x^2/2$. Conversely, $\phi(x) = (x, x^2)$ traces a parabola, the two directions remain visible, and moving θ costs in the whole plane. Figure 35 makes this progressive degeneration visible in a one-parameter family of feature maps. Figure 36 shows, at a fixed $\theta_\star$, what a step of the same length in each of the two eigen-directions does to the density: along the direction of the smallest eigenvalue, where the metric is nearly degenerate, the density barely moves, whereas along the direction of the largest eigenvalue the same step deforms it markedly. The anisotropy is thus read directly on the displacement function $\mathrm{d}p_\theta(\mathrm{d}\theta)$.

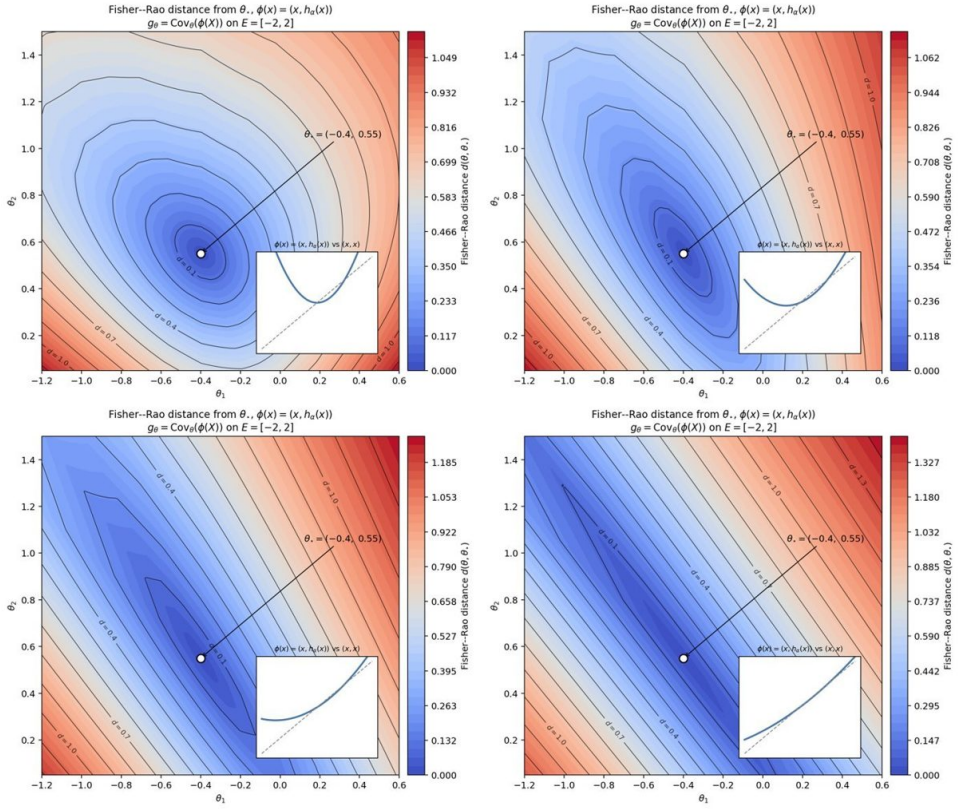


FIGURE 35. Progressive degeneration of the Fisher–Rao metric for a linear Gibbs family. The panels show the Fisher–Rao distance from the reference parameter $\theta_\star$ for

$$\phi_\alpha(x) = (x, h_\alpha(x)), \qquad h_\alpha(x) = (2-\alpha)x + (\alpha-1)x^2,$$

as α decreases from 2 towards 1. The metric is

$$g_\theta = \mathrm{Cov}_\theta\big(\phi_\alpha(X)\big).$$

For $\alpha = 2$, $\phi_2(x) = (x, x^2)$ traces a parabola: its two components are affinely independent and the metric is positive definite. As $\alpha \to 1$, the image of ϕ_α becomes increasingly close to the affine line $\phi_1(x) = (x, x)$. One eigenvalue of $\mathrm{Cov}_\theta(\phi_\alpha(X))$ then becomes small: the corresponding direction in parameter space produces an almost constant variation of the energy and therefore hardly changes the density. The progressive stretching of the iso-distance contours makes this near-degeneracy visible. In the limit $\alpha = 1$, the image is contained in a line, the parametrization becomes redundant, and the Fisher–Rao metric loses rank.

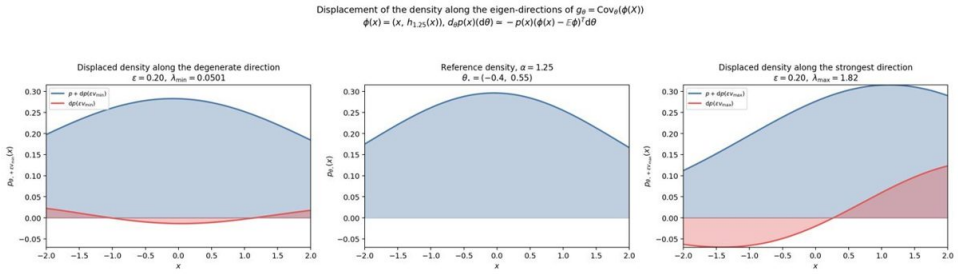


FIGURE 36. A step of the same Euclidean length ε along each eigen-direction of the metric at $\theta_\star$, for $\phi(x) = (x, h_{1.25}(x))$. The middle panel shows the reference density $p_{\theta_\star}$; the side panels show the displaced density $p_{\theta_\star+\varepsilon v}$ in blue together with the displacement $\mathrm{d}p_{\theta_\star}(\varepsilon v)$ in red. On the left, the step is taken along the direction of the smallest eigenvalue: the displacement is weak, the density barely changes. On the right, the same step is taken along the direction of the largest eigenvalue: the displacement is much stronger and visibly reshapes the density.

Because g_θ varies with θ, the shortest paths in Θ are not Euclidean segments. How they bend is read on the derivatives of the metric. Since $g_{ij} = \partial_i \partial_j \log Z_\theta$ is a Hessian, these derivatives are totally symmetric, and

$$\partial_i \partial_j \partial_k \log Z_\theta = -\mathbb{E}_\theta\big[(\phi_i - \mathbb{E}_\theta \phi_i)(\phi_j - \mathbb{E}_\theta \phi_j)(\phi_k - \mathbb{E}_\theta \phi_k)\big] .$$

The sign comes from the parametrization $e^{-\phi^\top \theta}$: differentiating the density once more brings down a factor $-(\phi - \mathbb{E}\phi)$, whence the centered moment of order three. When this third moment is large, $\mathrm{Cov}_\theta(\phi)$ changes quickly along a path, and the geodesic leaves the Euclidean chord.

Figure 37 shows a geodesic between two points of $\Theta \subset \mathbb{R}^2$ for two examples on $E \subset \mathbb{R}$, $\phi(x) = (x, x^2)$ and $\phi(x) = (x^2, \cos x)$, then for the trigonometric example on $E \subset \mathbb{R}^2$. At the top, the path in parameter space (in red), compared with the Euclidean chord (in dashes); at the bottom, the densities p_θ along the trajectory. The first path clearly leaves the line: the two components of $\phi = (x, x^2)$ remain visible. In the two other cases, the gap is more subtle, but the evolution of the densities remains clearly visible.

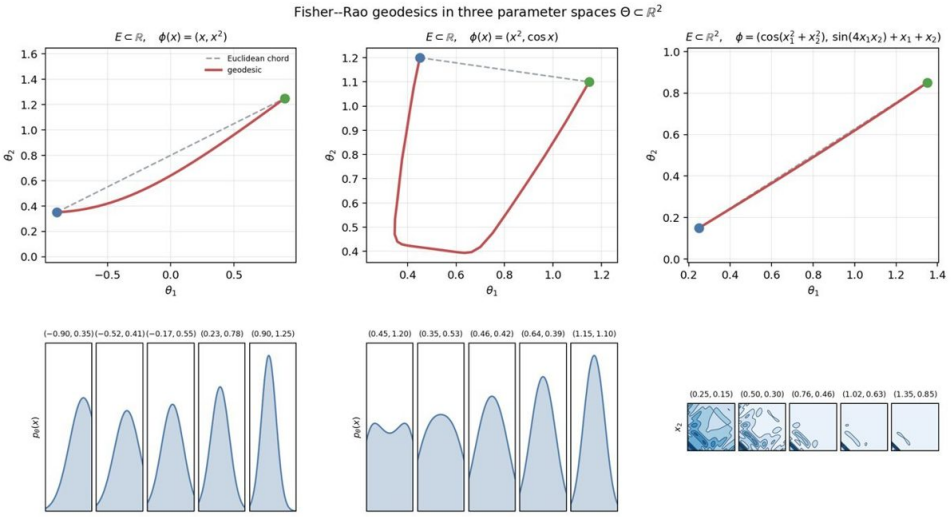


FIGURE 37. Numerical approximations of Fisher–Rao geodesics (discrete energy) on three parameter spaces $\Theta \subset \mathbb{R}^2$, the metric $G(\theta) = \mathrm{Cov}_\theta(\phi)$ being evaluated by a Riemann sum on a bounded window. The first two families are defined on $E \subset \mathbb{R}$, the third on $E \subset \mathbb{R}^2$. Departure in blue, arrival in green; densities at the intermediate points.

7. Non-linear Gibbs family

The next step is to let the energy depend on θ in a non-linear way. The densities take the form

$$p_\theta(x) = \frac{1}{Z_\theta}\, e^{-U(\theta,x)}, \qquad Z_\theta = \int_E e^{-U(\theta,x)}\, dx,$$

where $U\colon \Theta \times E \to \mathbb{R}$ is measurable in x for each θ and of class C^3 in θ. We assume moreover that U is sufficiently integrable, locally uniformly in θ, so that all differentiations under the integral sign and expectations appearing below are justified. The linear family of the previous section corresponds to the case

$$U(\theta, x) = \phi(x)^\top \theta.$$

The Gaussian also belongs to this nonlinear Gibbs setting:

$$U(\mu, \sigma; x) = \frac{(x-\mu)^2}{2\sigma^2}$$

is not linear in (μ, σ).

Two quick examples (figure 38). In dimension one, $U(\theta, x) = \cos(x\theta_1 + x^2\theta_2)$. In dimension two, $U(\theta, x) = (x_1 - \theta_1^2)\theta_2 + x_2^2\theta_2$. The normalization constants are obtained by numerical integration on a bounded domain.

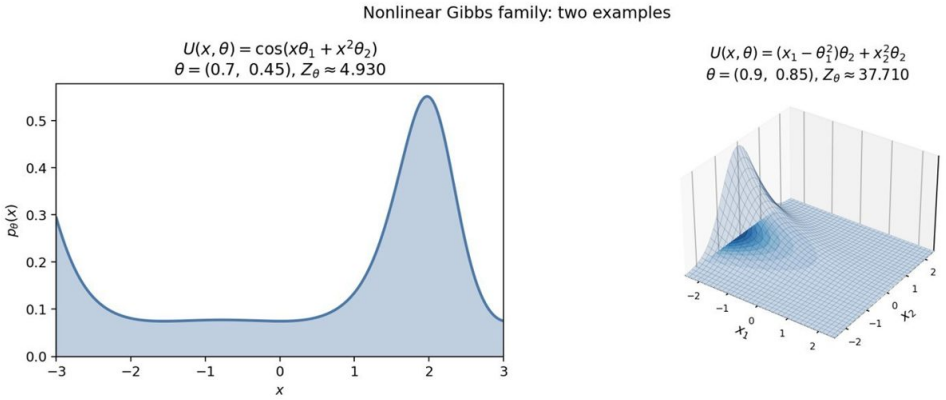


FIGURE 38. Two non-linear Gibbs densities. On the left, $E \subset \mathbb{R}$ and $U(\theta, x) = \cos(x\theta_1 + x^2\theta_2)$; on the right, $E \subset \mathbb{R}^2$ and $U(\theta, x) = (x_1 - \theta_1^2)\theta_2 + x_2^2\theta_2$.

7.1. Differential

Let us differentiate $p_\theta(x) = \dfrac{1}{Z_\theta} e^{-U(\theta,x)}$. Since Θ is an open subset of $\mathbb{R}^k$, we have $T_\theta\Theta \simeq \mathbb{R}^k$. For $d\theta \in T_\theta\Theta \simeq \mathbb{R}^k$,

$$dp_\theta(d\theta)(x) = -\left(\frac{dZ_\theta(d\theta)}{Z_\theta^2} + \frac{d_\theta U(\theta,x)(d\theta)}{Z_\theta}\right) e^{-U(\theta,x)}.$$

Factoring out $1/Z_\theta$ gives

$$dp_\theta(d\theta) = -p_\theta\left(\frac{dZ_\theta(d\theta)}{Z_\theta} + d_\theta U(d\theta)\right).$$

Now $dZ_\theta(d\theta) = -\int_E d_\theta U\, e^{-U}\, dx$, so $dZ_\theta(d\theta)/Z_\theta = -\mathbb{E}_\theta[d_\theta U(d\theta)]$, and

$$dp_\theta(d\theta) = -p_\theta\big(d_\theta U(d\theta) - \mathbb{E}_\theta[d_\theta U(d\theta)]\big).$$

The integral in x vanishes: $dp_\theta(d\theta)$ is indeed tangent to $\mathcal{P}_+(E)$. In coordinates, $d_\theta U(d\theta) = \sum_i d\theta^i\, \partial_i U$ and

$$\partial_i p_\theta = -p_\theta\big(\partial_i U - \mathbb{E}_\theta[\partial_i U]\big).$$

7.2. Metric

The Fisher–Rao metric on the densities is $g^{\mathrm{FR}}_p(u,v) = \int_E \frac{u\,v}{p}\, dx$. The pullback onto Θ is written

$$g_\theta(d\theta_1, d\theta_2) = g^{\mathrm{FR}}_{p_\theta}\big(dp_\theta(d\theta_1),\, dp_\theta(d\theta_2)\big) = \int_E \frac{dp_\theta(d\theta_1)\, dp_\theta(d\theta_2)}{p_\theta}\, dx.$$

Substituting the formula for dp_θ, the factors p_θ cancel and we are left with

$$\begin{aligned} g_\theta(d\theta_1, d\theta_2) &= \mathbb{E}_\theta\big[\big(d_\theta U(d\theta_1) - \mathbb{E}_\theta[d_\theta U(d\theta_1)]\big)\big(d_\theta U(d\theta_2) - \mathbb{E}_\theta[d_\theta U(d\theta_2)]\big)\big] \\ &= \mathrm{Cov}_\theta\big(d_\theta U(d\theta_1),\, d_\theta U(d\theta_2)\big). \end{aligned}$$

In short,

$$\boxed{g_\theta(d\theta_1, d\theta_2) = \mathrm{Cov}_\theta\big(d_\theta U(d\theta_1),\, d_\theta U(d\theta_2)\big)}.$$

In coordinates,

$$\boxed{g_{ij}(\theta) = \mathrm{Cov}_\theta(\partial_i U,\, \partial_j U) = \mathbb{E}_\theta[\partial_i U\, \partial_j U] - \mathbb{E}_\theta[\partial_i U]\, \mathbb{E}_\theta[\partial_j U]}.$$

In matrix form, if we write $\nabla_\theta U = (\partial_1 U, \dots, \partial_k U)$ for the gradient of U in θ, the matrix of the metric is simply the covariance of this gradient, when X follows p_θ:

$$g_\theta(d\theta_1, d\theta_2) = d\theta_1^\top \,\mathrm{Cov}_\theta\big(\nabla_\theta U(\theta, X)\big)\, d\theta_2.$$

In the linear case $U(\theta, x) = \phi(x)^\top \theta$, we have $\nabla_\theta U = \phi(x)$ and we recover $g_\theta(d\theta_1, d\theta_2) = d\theta_1^\top \mathrm{Cov}_\theta(\phi)\, d\theta_2$.

7.3. Analysis of the metric

The previous formula reads directly. For $d\theta \in T_\theta\Theta \simeq \mathbb{R}^k$,

$$g_\theta(d\theta, d\theta) = \mathrm{Var}_\theta\big(\nabla_\theta U(\theta, X)^\top d\theta\big).$$

The cost of an infinitesimal displacement $d\theta$ is therefore the variance, under the current law p_θ, of the energy modification it produces. A parameter direction is costly when it modifies the energy very differently depending on x; it costs little when this modification stays almost constant over the states that p_θ gives weight to.

That this is a variance, and not a second-order moment, has a simple reading. Adding to the energy a quantity independent of x does not change the density: it is absorbed by the normalization constant. The metric therefore sees only the fluctuations of the energy variation around its mean. In particular, a direction $d\theta$ such that

$$\nabla_\theta U(\theta, x)^\top d\theta = c$$

for almost every x, with c independent of x, has zero cost: to first order, it only adds a constant to the energy and does not move p_θ in the space of densities.

We also specify when the pullback actually defines a Riemannian metric. The matrix $\mathrm{Cov}_\theta\big(\nabla_\theta U(\theta, X)\big)$ is always symmetric positive semidefinite, as soon as the second-order moments exist. It is positive definite if and only if

$$\forall\, d\theta \neq 0, \qquad \mathrm{Var}_\theta\big(\nabla_\theta U(\theta, X)^\top d\theta\big) > 0,$$

that is, if no non-trivial linear combination of the components of $\nabla_\theta U(\theta, \cdot)$ is constant almost everywhere. This is exactly the injectivity of

$$dp_\theta \colon T_\theta\Theta \longrightarrow T_{p_\theta}\mathcal{P}_+(E).$$

In other words, the covariance is positive definite precisely when $\theta \mapsto p_\theta$ is an immersion at θ. A degeneracy reflects a redundancy of the parametrization: there exists a non-zero direction in the parameter space that does not modify the distribution to first order.

The passage from the linear case to the non-linear case can also be read in the normalization constant. We have

$$\nabla_\theta \log Z_\theta = -\mathbb{E}_\theta\big[\nabla_\theta U(\theta, X)\big],$$

and, differentiating a second time,

$$\mathrm{Hess}_\theta \log Z_\theta = \mathrm{Cov}_\theta\big(\nabla_\theta U(\theta, X)\big) - \mathbb{E}_\theta\big[\mathrm{Hess}_\theta\, U(\theta, X)\big].$$

The Fisher matrix therefore reads

$$\boxed{\operatorname{Cov}_\theta\big(\nabla_\theta U(\theta, X)\big) = \operatorname{Hess}_\theta \log Z_\theta + \mathbb{E}_\theta\big[\operatorname{Hess}_\theta U(\theta, X)\big]}.$$

In the linear family $U(\theta, x) = \phi(x)^\top \theta$, the second term vanishes and we recover the Hessian metric

$$\operatorname{Cov}_\theta\big(\phi(X)\big) = \operatorname{Hess}_\theta \log Z_\theta.$$

In the non-linear case, the covariance remains the fundamental quantity, but the metric is no longer in general the Hessian of $\log Z_\theta$ alone: an additional term appears, which measures the curvature of the energy with respect to the parameters.

8. **Push-forward family**

For this last family, we reverse the point of view. So far the family was given by a formula for the density, and θ tuned this formula. We now fix a base law once and for all, and it is the observation space that gets deformed.

Let $q \in \mathcal{P}_+(E)$ be fixed and $\phi_\theta \colon E \to E$ a family of $\mathcal{C}^2$ diffeomorphisms, indexed by θ in an open set $\Theta \subset \mathbb{R}^k$, with $\theta \mapsto \phi_\theta$ differentiable. Throughout this section, x is the variable of E on which the base law q is read, and $y = \phi_\theta(x)$ the variable on which p_θ is read. The parametric family is the image of q under the transport:

$$p_\theta = (\phi_\theta)_\sharp\, q, \qquad p_\theta(y) = \frac{q(\phi_\theta^{-1}(y))}{\left|\det \dfrac{\partial \phi}{\partial x}(\theta,\, \phi_\theta^{-1}(y))\right|},$$

by the change-of-variables formula, where $\frac{\partial \phi}{\partial x}(\theta, x)$ denotes the Jacobian matrix of ϕ_θ at the point x; the notation for the derivatives of ϕ is fixed in the next subsection. Read at the transported point $y = \phi_\theta(x)$ rather than at y itself, the same formula involves only ϕ_θ:

$$p_\theta(\phi_\theta(x)) \left|\det \frac{\partial \phi}{\partial x}(\theta, x)\right| = q(x), \qquad x \in E.$$

This is the form we shall work with. Watch the notation: ϕ denoted the sufficient statistic of the linear Gibbs family, with values in $\mathbb{R}^k$; here ϕ_θ transports E into itself, and it is θ that indexes it.

The affine case is already known. With $\phi_\theta(x) = \mu + Ax$, $A = R\,\operatorname{diag}(\sigma)$, the base $q = \mathcal{N}(0, I_n)$ gives the multivariate Gaussian [12] and the standard Laplace base gives the elliptical Laplace [19]. The two families of the previous sections are therefore push-forwards, with affine transport. What follows amounts to asking what happens to the geometry when we let ϕ_θ bend.

Three nonlinear examples (figure 39), all with q a standard Gaussian. In dimension one, $\phi_\theta(x) = \theta_1 x + \theta_2 x^3$ with $\theta_1, \theta_2 > 0$, and $\phi_\theta(x) = x + \theta_1 \sin(\theta_2 x)$ under $|\theta_1\theta_2| < 1$; in both cases the condition on θ ensures strict monotonicity, hence the diffeomorphism, and defines Θ. In dimension two, $\phi_\theta(x) = (x_1 + \theta_1 \tanh x_2,\ \theta_2 x_2)$ with $\theta_2 > 0$: the Jacobian is triangular, with determinant θ_2.

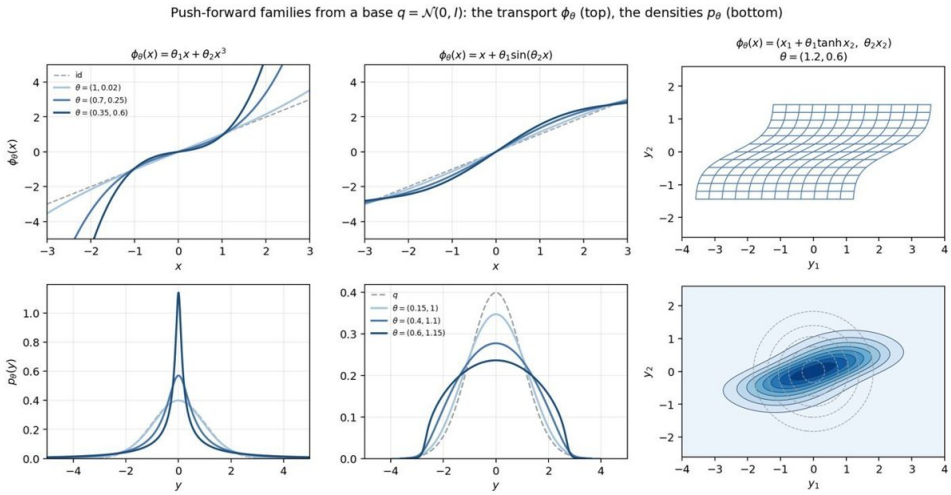


FIGURE 39. Three push-forward families of a standard Gaussian base. At the top, the transport ϕ_θ itself: the curves $x \mapsto \phi_\theta(x)$ for the two one-dimensional examples (the identity in dashes), and the image of a grid for the two-dimensional example. At the bottom, the resulting densities p_θ, with the base q in dashes.

8.1. **Differential**

The computation below is the one used for the Reynolds transport theorem: we work in the initial coordinates, we write that the probability carried by a transported region does not change, and we differentiate.

8.1.1. **Notation for the two variables**

Two variables are in play, the parameter and the point of E, and the whole computation consists in keeping track of which one is being differentiated. We therefore regard the transport as a function of both,

$$\phi \colon \Theta \times E \to E, \qquad (\theta, x) \longmapsto \phi(\theta, x) = \phi_\theta(x),$$

where x is the point read by q and $y = \phi(\theta, x)$ the point read by p_θ. Its two partial derivatives are written

$$\frac{\partial \phi}{\partial x}(\theta, x) \in M_n(\mathbb{R}) \quad \text{derivative in the space variable, } \theta \text{ fixed,}$$
$$\frac{\partial \phi}{\partial \theta}(\theta, x) \in M_{n,k}(\mathbb{R}) \quad \text{derivative in the parameter, } x \text{ fixed,}$$

both in the canonical bases, so that $\frac{\partial \phi}{\partial \theta}(\theta, x)\, d\theta \in \mathbb{R}^n$ is the variation of the transported point produced by the parameter variation $d\theta \in \mathbb{R}^k$. The same convention is used for any function of two variables: $\frac{\partial p}{\partial \theta}(\theta, y)\, d\theta$ is the variation of the density at a fixed point y, and it is the object we are after, written $dp_\theta(d\theta)$ in the previous sections. For a function of the space variable alone, such as the velocity field v below, we write $\frac{\partial v}{\partial y}$ for its Jacobian matrix,

and ∇, div for its gradient and divergence. Since Θ is an open subset of $\mathbb{R}^k$, we still have $T_\theta\Theta \simeq \mathbb{R}^k$, and a tangent direction is simply a vector $d\theta \in \mathbb{R}^k$.

8.1.2. **The velocity field of the deformation**

Fix θ and a direction $d\theta$, and let

$$\theta(s) = \theta + s\,d\theta, \qquad s \in \,]-\epsilon, \epsilon[\,,$$

a function of s with values in Θ, with $\theta(0) = \theta$ and $\theta'(s) = d\theta$. The maps $\phi_{\theta(s)}$ then form a one-parameter family of diffeomorphisms of E: this is a flow, with s in place of time. The trajectory of an initial point x is the function $s \mapsto \phi(\theta(s), x)$, and the chain rule gives its velocity,

$$\frac{d}{ds}\,\phi(\theta(s), x) = \frac{\partial\phi}{\partial\theta}(\theta(s), x)\,d\theta.$$

As for any flow, this velocity can be read as a field on the space where the points currently are. We define $v_{d\theta}\colon E \to \mathbb{R}^n$ by

$$\frac{\partial\phi}{\partial\theta}(\theta, x)\,d\theta = v_{d\theta}(\phi(\theta, x)) \qquad \text{for all } x \in E,$$

the relation $\partial_\theta\phi = v \circ \phi$, which is the analogue of $\dot\gamma(t) = v(\gamma(t))$ for an ordinary flow. Since ϕ_θ is a bijection, this determines $v_{d\theta}$. Taking $d\theta = e_i$ gives the coordinate fields v_i, characterized by $\frac{\partial\phi}{\partial\theta_i}(\theta, x) = v_i(\phi(\theta, x))$. The field $v_{d\theta}$ lives on E; it is not a tangent vector to Θ.

Figure 40 makes the relation visible for the two-dimensional example $\phi_\theta(x) = (x_1 + \theta_1 \tanh x_2,\ \theta_2 x_2)$, at $\theta = (0{,}8,\ 1{,}2)$ and in the direction $d\theta = e_1$. The arrows are the field $v_{d\theta}$ read at the current positions $y = \phi_\theta(x)$; they are computed from $\frac{\partial\phi}{\partial\theta}(\theta, x)\,d\theta = (\tanh x_2,\ 0)$, without inverting ϕ_θ. The dashed grid is the same material grid after the finite step $\theta + \varepsilon d\theta$, with $\varepsilon = 0{,}15$: it is the integrated displacement that the velocity field describes infinitesimally.

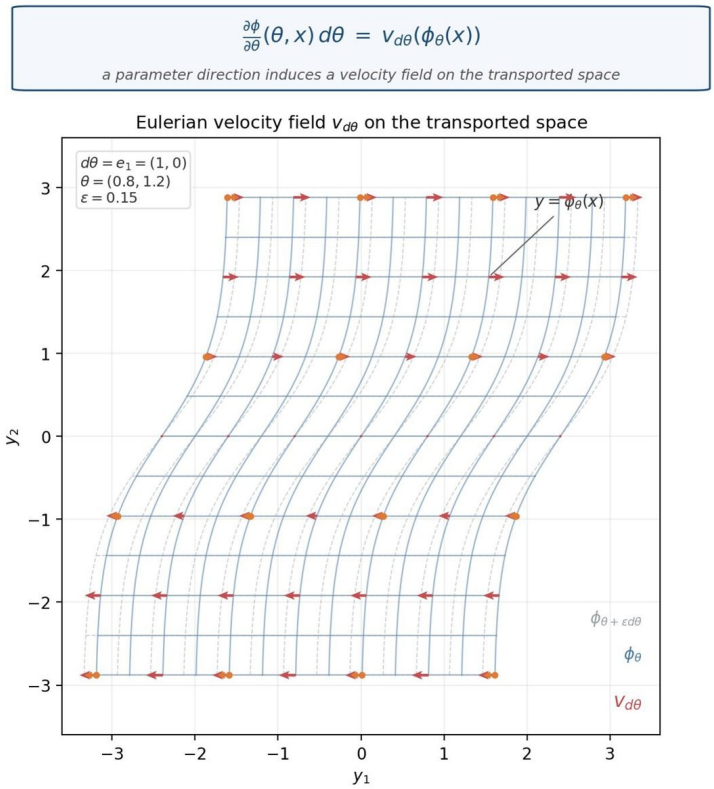


FIGURE 40. Eulerian velocity field induced by a parameter direction. Transport $\phi_\theta(x) = (x_1 + \theta_1 \tanh x_2,\ \theta_2 x_2)$ at $\theta = (0{,}8,\ 1{,}2)$, with $d\theta = e_1$. Solid: the grid transported by ϕ_θ. Dashed: the same grid after $\theta + \varepsilon d\theta$, $\varepsilon = 0{,}15$. Red arrows: $v_{d\theta}(y) = \frac{\partial \phi}{\partial \theta}(\theta, x)\, d\theta$ at $y = \phi_\theta(x)$, computed from the initial point x. A direction in Θ becomes a vector field on E.

8.1.3. **Conservation of probability**

Let $A \subseteq E$ be a Borel set of initial points and $A_s = \phi_{\theta(s)}(A)$ the region it occupies once transported. Since $\phi_{\theta(s)}$ is a bijection, the initial points that land in A_s are exactly those of A, so the probability carried by A_s does not depend on s:

$$\int_{A_s} p_{\theta(s)}(y)\, dy = \int_A q(x)\, dx \qquad \text{for all } s.$$

The change of variables $y = \phi(\theta(s), x)$ brings the left-hand side back to the initial coordinates,

$$\int_A p_{\theta(s)}(\phi(\theta(s), x)) \left| \det \frac{\partial \phi}{\partial x}(\theta(s), x) \right| dx = \int_A q(x)\, dx.$$

Both integrals are now over the same fixed region A, and A is arbitrary, so the integrands agree everywhere:

$$p_{\theta(s)}(\phi(\theta(s), x)) \left| \det \frac{\partial \phi}{\partial x}(\theta(s), x) \right| = q(x) \qquad \text{for all } x \in E \text{ and all } s.$$

This is the change-of-variables formula of the introduction to this section. We call it the transport identity, and everything follows by differentiating it with respect to s.

Figure 41 illustrates the identity at a fixed θ. The base is the standard Gaussian $q = \mathcal{N}(0, I_2)$, and the transport is the two-dimensional example of the introduction, $\phi_\theta(x) = (x_1 + a \tanh x_2,\, bx_2)$ with $a = 0{,}8$ and $b = 1{,}3$. A material region A is deformed into $A_\theta = \phi_\theta(A)$, the density becomes $p_\theta = (\phi_\theta)_\sharp q$, and the same particles that sat in A are recovered in A_θ. The probability mass they carry does not change:

$$\int_A q(x)\,dx = \int_{\phi_\theta(A)} p_\theta(y)\,dy.$$

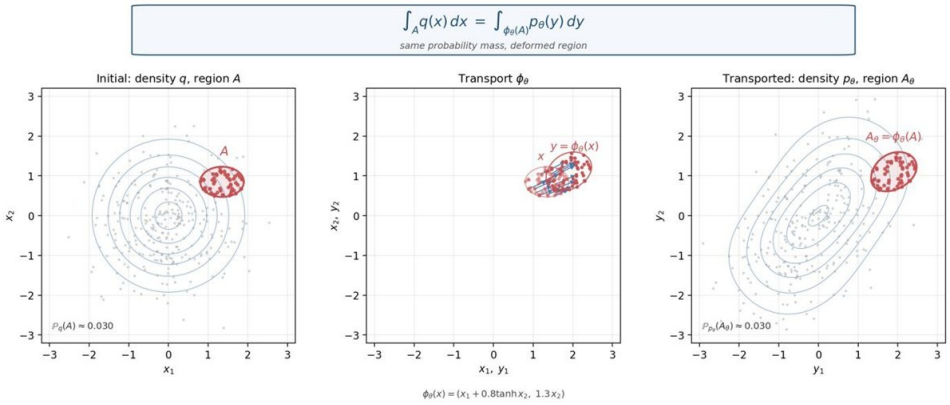


FIGURE 41. Conservation of probability under a push-forward. Base $q = \mathcal{N}(0, I_2)$ and transport $\phi_\theta(x) = (x_1 + 0{,}8 \tanh x_2,\, 1{,}3\, x_2)$, with $\det \frac{\partial \phi}{\partial x} = 1{,}3$. Left: density q, region A, and particles from the base (in red those of A). Middle: the same particles of A mapped by ϕ_θ. Right: density $p_\theta = (\phi_\theta)_\sharp q$, transported region $A_\theta = \phi_\theta(A)$, and the same particles after transport. The probability mass of the material region is unchanged.

8.1.4. **Rate of change of the volume element**

One ingredient is needed first: the derivative of the Jacobian determinant. As in the multivariate Gaussian section, for invertible M one has

$$d(\det)(M)(H) = \det(M)\ \mathrm{tr}(M^{-1}H).$$

Composing with $\log|\cdot|$ cancels the factor $\det(M)$ and leaves the trace. It remains to differentiate the matrix $\frac{\partial \phi}{\partial x}(\theta(s), x)$ with respect to s, at fixed x. The chain rule in s produces a mixed second derivative of ϕ, and the two derivatives may be exchanged:

$$\frac{d}{ds}\,\frac{\partial \phi}{\partial x}(\theta(s), x) = \frac{\partial^2 \phi}{\partial \theta\, \partial x}(\theta(s), x)\,d\theta = \frac{\partial}{\partial x}\left(\frac{\partial \phi}{\partial \theta}(\theta(s), x)\,d\theta\right),$$

so what we have to compute is the space derivative of the velocity of the trajectories. At $s = 0$ this velocity is $v_{d\theta} \circ \phi_\theta$ by the definition of $v_{d\theta}$, and the chain rule in x gives

$$\frac{d}{ds}\bigg|_{s=0} \frac{\partial \phi}{\partial x}(\theta(s), x) = \frac{\partial v_{d\theta}}{\partial y}(\phi(\theta, x)) \, \frac{\partial \phi}{\partial x}(\theta, x).$$

Putting the two together,

$$\frac{d}{ds}\bigg|_{s=0} \log\left|\det \frac{\partial \phi}{\partial x}(\theta(s), x)\right| = \operatorname{tr}\left(\frac{\partial \phi}{\partial x}(\theta, x)^{-1} \, \frac{\partial v_{d\theta}}{\partial y}(\phi(\theta, x)) \, \frac{\partial \phi}{\partial x}(\theta, x)\right).$$

The trace is invariant under conjugation, so the two factors $\frac{\partial \phi}{\partial x}(\theta, x)$ cancel and there remains

$$\operatorname{tr} \frac{\partial v_{d\theta}}{\partial y}(\phi(\theta, x)) = \operatorname{div} v_{d\theta}(\phi(\theta, x)).$$

The rate at which the volume element is stretched along a trajectory is the divergence of the velocity field at the current position, the same statement as in a compressible fluid.

Figure 42 isolates this geometric content. For a velocity field v on E, a small material cell A is moved by the first-order map $y \mapsto y + \varepsilon v(y)$. Positive divergence enlarges the cell, negative divergence shrinks it, and a pure shear ($\operatorname{div} v = 0$) changes the shape while preserving the area to first order. What $\operatorname{div} v$ records is therefore the infinitesimal rate of volume change, not the deformation of the shape itself:

$$\frac{d}{ds} \log(\mathrm{Vol}) = \operatorname{div} v.$$

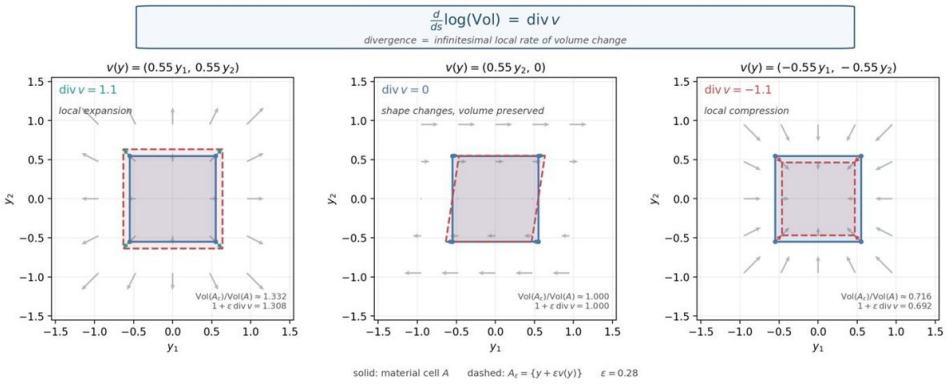


FIGURE 42. Divergence as infinitesimal local rate of volume change. The same material cell A (solid) is deformed by $y \mapsto y + \varepsilon v(y)$ (dashed), $\varepsilon = 0{,}28$. Left: radial expansion $v(y) = (ay_1, ay_2)$, $\operatorname{div} v = 2a > 0$. Middle: shear $v(y) = (ay_2, 0)$, $\operatorname{div} v = 0$: the cell becomes a parallelogram, the area is unchanged. Right: radial compression $v(y) = (-ay_1, -ay_2)$, $\operatorname{div} v = -2a < 0$. The numerical ratios $\mathrm{Vol}(A_\varepsilon)/\mathrm{Vol}(A)$ match $1 + \varepsilon \operatorname{div} v$ to first order.

8.1.5. **Differentiating**

Take the logarithm of the transport identity:

$$\log p_{\theta(s)}\big(\phi(\theta(s),x)\big) + \log\left|\det \frac{\partial \phi}{\partial x}(\theta(s),x)\right| = \log q(x),$$

and differentiate at $s=0$ with x fixed. The right-hand side does not depend on s, so its derivative vanishes. In the first term both slots move, the parameter and the evaluation point, so the chain rule gives two contributions, the velocity of the evaluation point being $v_{d\theta}(y)$ by definition of the velocity field:

$$\left.\frac{d}{ds}\right|_{s=0} \log p_{\theta(s)}\big(\phi(\theta(s),x)\big) = \frac{\partial \log p}{\partial \theta}(\theta,y)\,d\theta + \langle \nabla \log p_\theta(y),\, v_{d\theta}(y)\rangle\,,$$

where $y=\phi(\theta,x)$. The second term is $\operatorname{div} v_{d\theta}(y)$, by the computation above. Adding the two and setting the sum to zero,

$$\boxed{\frac{\partial \log p}{\partial \theta}(\theta,y)\,d\theta = -\,\langle \nabla \log p_\theta(y),\, v_{d\theta}(y)\rangle - \operatorname{div} v_{d\theta}(y)}$$

and since ϕ_θ maps E onto E, this holds at every point y. The left-hand side plus the inner product is the material derivative of $\log p$ along the flow: the identity says that this material derivative is $-\operatorname{div} v_{d\theta}$.

Multiply now by $p_\theta(y)$, using $p_\theta \nabla \log p_\theta = \nabla p_\theta$ and the product rule

$$\operatorname{div}\big(p_\theta v_{d\theta}\big) = \langle \nabla p_\theta,\, v_{d\theta}\rangle + p_\theta \operatorname{div} v_{d\theta}.$$

There remains

$$\boxed{dp_\theta(d\theta) = \frac{\partial p}{\partial \theta}(\theta,\cdot)\,d\theta = -\operatorname{div}\big(p_\theta\, v_{d\theta}\big).}$$

This is the continuity equation of the flow $\big(\phi_{\theta(s)}\big)_s$. Along the curve $\theta(s)$,

$$\left.\frac{d}{ds}\right|_{s=0} p_{\theta(s)} + \operatorname{div}\big(p_\theta\, v_{d\theta}\big) = 0,$$

with momentum, or mass flux, $p_\theta v_{d\theta}$. The deformation of the density is thus $u_{d\theta} = -\operatorname{div}(p_\theta v_{d\theta})$: only the divergence of the momentum is seen, so other velocity fields produce the same $u_{d\theta}$. The tangency to $\mathcal{P}_+(E)$ is immediate, $\int_E \operatorname{div}(p_\theta v_{d\theta})\,dy = 0$ by the divergence theorem, as soon as $p_\theta v_{d\theta}$ decays fast enough at infinity. In coordinates,

$$\frac{\partial p}{\partial \theta_i}(\theta,\cdot) = -\operatorname{div}\big(p_\theta\, v_i\big).$$

Figure 43 isolates the local mechanism. A material region carries a fixed probability mass; the field $v_{d\theta}$ expands or compresses that region. With the mass unchanged, the density p_θ must move in the opposite direction: lighter under expansion ($\operatorname{div} v_{d\theta} > 0$), darker under compression ($\operatorname{div} v_{d\theta} < 0$). This is the content of

$$dp_\theta(d\theta) = -\operatorname{div}\big(p_\theta\, v_{d\theta}\big).$$

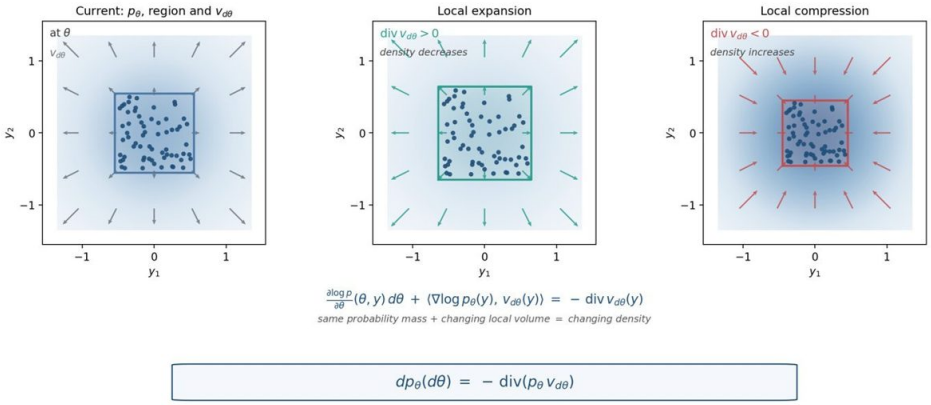


FIGURE 43. From volume change to density change, under conservation of probability. Left: a material cell at θ, with local shading of p_θ, particles for the probability mass, and arrows of $v_{d\theta}$. Middle: local expansion $\operatorname{div} v_{d\theta} > 0$: the same particles occupy a larger cell, the density decreases. Right: local compression $\operatorname{div} v_{d\theta} < 0$: the same particles occupy a smaller cell, the density increases. The particle count is identical in the three panels.

8.2. Metric

Set

$$\xi_{d\theta} = \frac{\operatorname{div}(p_\theta v_{d\theta})}{p_\theta} = \langle \nabla \log p_\theta,\, v_{d\theta} \rangle + \operatorname{div} v_{d\theta}.$$

From the continuity equation of the previous subsection, $dp_\theta(d\theta) = -\operatorname{div}(p_\theta v_{d\theta})$, so $\xi_{d\theta} = -dp_\theta(d\theta)/p_\theta$. The pullback of $g_p^{\mathrm{FR}}(u,w) = \int_E \frac{u\,w}{p}\, dy$ is then

$$g_\theta(d\theta_1, d\theta_2) = \int_E \frac{\operatorname{div}(p_\theta v_{d\theta_1})\, \operatorname{div}(p_\theta v_{d\theta_2})}{p_\theta}\, dy = \mathbb{E}_\theta\big[\xi_{d\theta_1}\, \xi_{d\theta_2}\big].$$

The random variable $\xi_{d\theta}$ has mean zero, but not because $\operatorname{div}(p_\theta v_{d\theta})$ vanishes pointwise. Each p_θ is a probability density, $\int_E p_\theta(y)\, dy = 1$. Differentiating the normalization in the direction $d\theta$ gives $\int_E dp_\theta(d\theta)\, dy = 0$. Substituting the continuity equation,

$$\mathbb{E}_\theta[\xi_{d\theta}] = \int_E \xi_{d\theta}\, p_\theta\, dy = \int_E \operatorname{div}(p_\theta v_{d\theta})\, dy = 0.$$

This is the tangent-space condition $\int_E u = 0$ of $\mathcal{P}_+(E)$, inherited from the fact that the total mass remains one. The product of two centered variables is a covariance, taken when Y follows p_θ:

$$\boxed{g_\theta(d\theta_1, d\theta_2) = \operatorname{Cov}_\theta(\xi_{d\theta_1},\, \xi_{d\theta_2}).}$$

We recover the form of the previous sections, with ξ in place of the sufficient statistic or of $\nabla_\theta U$. In coordinates the score is $\frac{\partial \log p}{\partial \theta_i} = -\xi_i$, hence

$$g_{ij}(\theta) = \mathbb{E}_\theta[\xi_i \xi_j] = \mathbb{E}_\theta\left[\frac{\partial \log p_\theta}{\partial \theta_i} \frac{\partial \log p_\theta}{\partial \theta_j}\right],$$

the usual Fisher information. The pullback is automatically positive semi-definite. It is a Riemannian metric when $\theta \mapsto p_\theta$ is an immersion, equivalently when $dp_\theta(d\theta) = 0$ implies $d\theta = 0$. The density only sees $-\operatorname{div}(p_\theta v_{d\theta})$, so a nonzero velocity field may produce no first-order change of p_θ.

One last expression, the one used for the numerical computation. The change of variables $y = \phi(\theta, x)$ brings the expectation back to the initial coordinates:

$$g_{ij}(\theta) = \mathbb{E}_{X\sim q}\big[\xi_i(\phi(\theta, X))\, \xi_j(\phi(\theta, X))\big].$$

The divergence term is the trace computed above, with $d\theta = e_i$. For the gradient term, differentiate the logarithm of the transport identity with respect to x at fixed θ, which gives the score read in the initial coordinates:

$$w(x) = \nabla\left(\log q - \log\left|\det \frac{\partial \phi}{\partial x}\right|\right)(x),$$
$$\nabla \log p_\theta(\phi(\theta, x)) = \left(\frac{\partial \phi}{\partial x}\right)^{-\top} w(x).$$

Omitting from now on the argument (θ, x) of the derivatives of ϕ, which are all taken at the same point,

$$\xi_i(\phi(\theta, x)) = \operatorname{tr}\left(\left(\frac{\partial \phi}{\partial x}\right)^{-1} \frac{\partial^2 \phi}{\partial x\, \partial \theta_i}\right) + \left\langle \left(\frac{\partial \phi}{\partial x}\right)^{-\top} w(x),\ \frac{\partial \phi}{\partial \theta_i}\right\rangle.$$

Everything is read on q and on the derivatives of ϕ at (θ, x). In a numerical implementation one solves linear systems against $\frac{\partial \phi}{\partial x}$ rather than forming the inverse explicitly.

8.3. **Analysis of the metric**

The divergence form says what the metric looks at. Fisher–Rao does not see $v_{d\theta}$, but $\operatorname{div}(p_\theta v_{d\theta})$: what costs is the local compression or dilation of mass, not the displacement itself. A direction $d\theta$ whose velocity field satisfies

$$\operatorname{div}(p_\theta\, v_{d\theta}) = 0$$

therefore costs nothing. The flux $p_\theta v_{d\theta}$ then circulates without creating or destroying mass anywhere: the points of E move, the density does not. Figure 44 shows a concrete instance: the isotropic Gaussian $p = \mathcal{N}(0, I_2)$ and the rotational field $v(y) = (-y_2, y_1)$. Then $\operatorname{div} v = 0$ and $\langle \nabla p, v\rangle = 0$, hence $\operatorname{div}(pv) = 0$ and $u = -\operatorname{div}(pv) = 0$.

These free directions are not an accident of the choice of examples. If $\sigma\colon E \to E$ preserves q, that is $\sigma_\sharp q = q$, then ϕ_θ and $\phi_\theta \circ \sigma$ give the same p_θ. The parametrization by transports is therefore massively redundant, and this redundancy must be broken for the metric to be positive definite. We already said it for the nonlinear Gibbs family: g_θ is positive definite exactly when $\theta \mapsto p_\theta$ is an immersion at θ. Here, the source of the degeneracy has a name: the rearrangements that preserve the base. The rotation of figure 44 is exactly such a rearrangement of the isotropic Gaussian.

This quotient viewpoint is closely related to the geometric description of spaces of densities through diffeomorphism groups; see Khesin, Lenells, Misioek, and Preston [6].

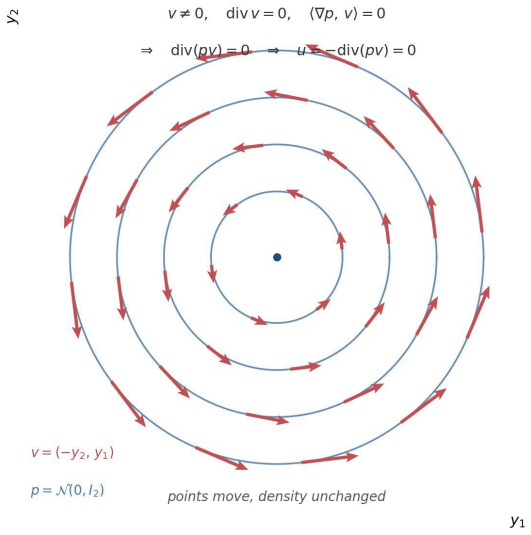


FIGURE 44. A nonzero velocity field invisible to Fisher–Rao. Level sets of $p = \mathcal{N}(0, I_2)$ and the rotational field $v(y) = (-y_2, y_1)$. The arrows are tangent to the contours: $\operatorname{div} v = 0$ and $\langle \nabla p, v \rangle = 0$, so $\operatorname{div}(pv) = 0$ and $u = -\operatorname{div}(pv) = 0$. The points move, the density does not change.

The affine case serves as a check. For a mean displacement, $v_{d\mu} = d\mu$ is constant, its divergence is zero, hence $\xi_{d\mu} = \langle \nabla \log p_\theta, d\mu \rangle$ and

$$g_\theta(d\mu, d\mu) = d\mu^\top \, \mathbb{E}_\theta\big[\nabla \log p_\theta \, \nabla \log p_\theta^\top\big] \, d\mu = d\mu^\top \Sigma^{-1} d\mu$$

for the Gaussian: the Mahalanobis distance of the Gaussian chapter, recovered without computing an integral. The dispersion block reads the same way, with the linear field $v(y) = dA\, A^{-1}(y - \mu)$, of constant divergence $\operatorname{tr}(dA\, A^{-1})$.

The identity $\partial_i \log p_\theta = -\xi_i$ deserves a pause: the score is nothing other than the local rate of mass compression produced by θ_i. Where the transport compresses, the log-density rises; where it dilates, it falls.

Figure 45 compares the two divergences. We start from $q = \mathcal{N}(0, I_2)$ and a transport that squeezes along x_1 twice, in the centre and in a tail:

$$\phi(x) = \big(x_1 + \alpha(\tanh x_1 + \tanh(x_1 - c)),\, x_2\big),$$

with $\alpha = -0{,}42$ and $c = 2{,}25$. At the bottom, a uniform grid and then its image: the two compressions are geometrically equivalent. At the top, $p = \phi_\sharp q$ is the same surface on both sides; the floor is $\operatorname{div} v$ on the left and $\operatorname{div}(pv)$ on the right, for the displacement $v = (\phi - \mathrm{id}) \circ \phi^{-1}$. The tail band, clear on the left, fades as soon as one weighs by p.

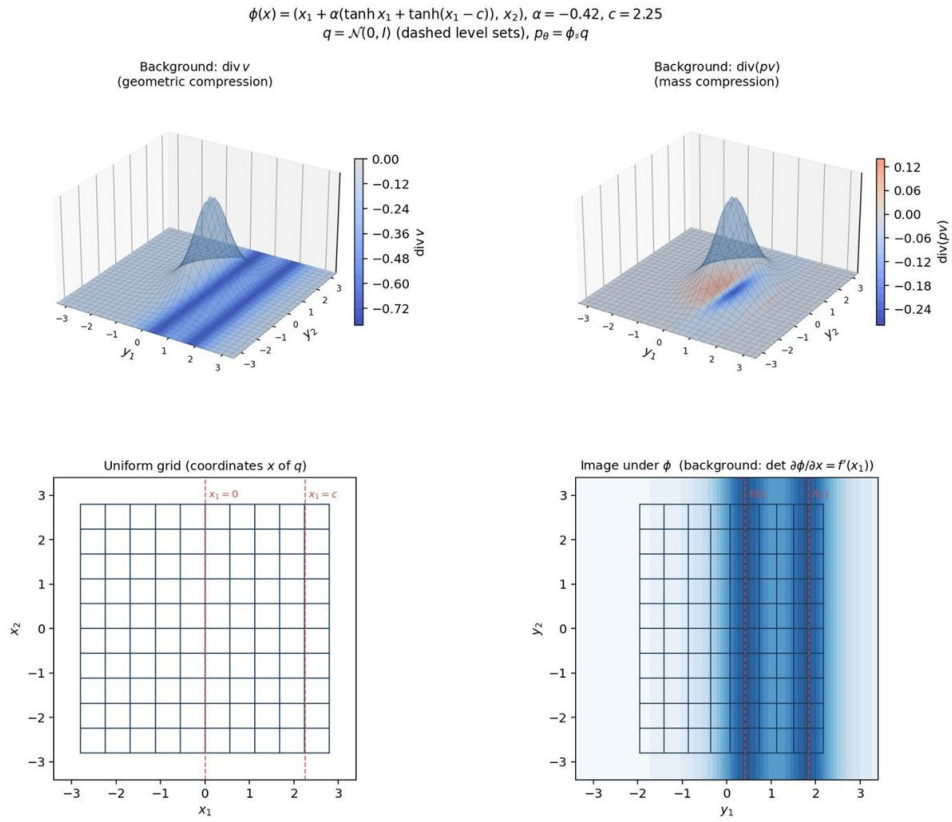


FIGURE 45. Transport $\phi(x) = (x_1 + \alpha(\tanh x_1 + \tanh(x_1 - c)), x_2)$ of a standard Gaussian, $\alpha = -0{,}42$, $c = 2{,}25$. At the top, the same density $p = \phi_\sharp q$; the floor is $\operatorname{div} v$ on the left and $\operatorname{div}(pv)$ on the right. At the bottom, the uniform grid in the coordinates of q, then its image by ϕ. The two squeezes read on the grid as clearly as each other; Fisher–Rao keeps only the one in the bulk.

We had opposed Fisher–Rao and Wasserstein in section 3.2. The tangent is $u = -\operatorname{div}(pv)$, and several v can realize it. For a given field, $\int_E \|v\|^2 p\, dy$ is its kinetic energy; the Otto norm of u is the infimum of these energies [8]. Fisher–Rao pays $\int_E \operatorname{div}(pv)^2/p\, dy$, which depends only on u. In dimension one, under suitable decay conditions at infinity, $u + \partial_y(pv) = 0$ determines the admissible velocity field uniquely, so its kinetic energy already realizes the Otto infimum. As soon as $n \geq 2$, the energy of $v_{d\theta}$ coming from ϕ_θ is only an upper bound a priori. A rigid translation remains the good example: it costs much to Wasserstein, which counts the displacement, and little to Fisher–Rao if the density is spread out, since the overlap remains good.

Figure 46 shows the geodesics for the two one-dimensional examples, whose parameter space is an open subset of $\mathbb{R}^2$. The metric is obtained by quadrature on the base, with the base-coordinate formula above; the geodesics minimize the discrete Riemannian energy with fixed endpoints. In both cases the path leaves the Euclidean chord. For the cubic family, θ_2

stays low longer than on the chord, then accelerates: digging the peak and stretching the tails is paid mainly at the end. For the sinusoidal family, it is the reverse, θ_2 takes the lead from the start, and the density flattens into a plateau before the amplitude θ_1 finishes its course.

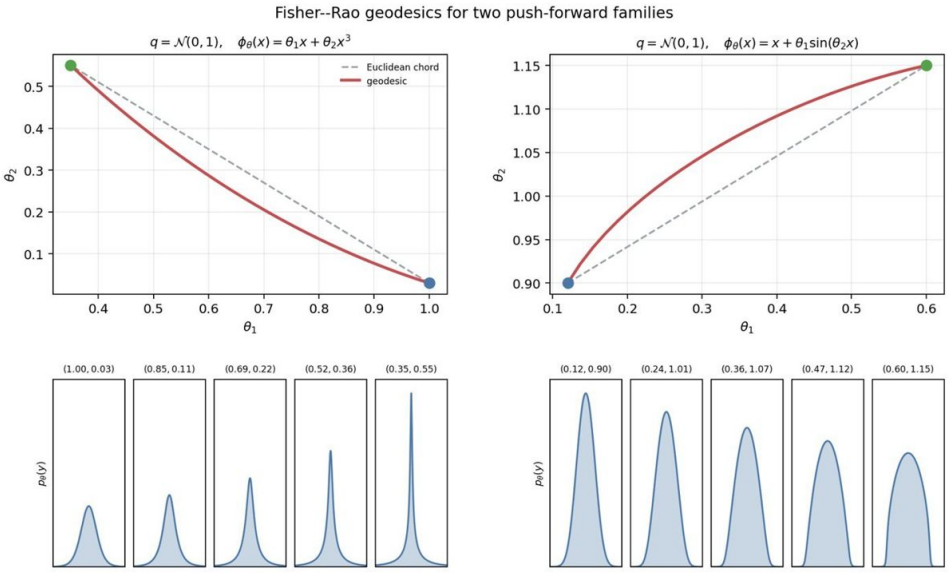


FIGURE 46. Numerical approximations of Fisher–Rao geodesics for two push-forward families, $q = \mathcal{N}(0,1)$ (discrete energy, metric by quadrature on the base). At the top the path in $\Theta \subset \mathbb{R}^2$ (in red), compared with the Euclidean chord (in dashes); at the bottom the densities p_θ along the path, with (θ_1, θ_2) in the title.

The two-dimensional example lends itself even better to reading, because the metric is explicit there. For $\phi_\theta(x) = (x_1 + \theta_1 \tanh x_2,\ \theta_2 x_2)$, the determinant $\det \frac{\partial \phi}{\partial x} = \theta_2$ does not depend on x, which cancels the gradient term of the log-Jacobian, and the scores reduce to

$$\xi_1(\phi(\theta,x)) = -x_1 \tanh x_2, \qquad \xi_2(\phi(\theta,x)) = \frac{1 + \theta_1(1-\tanh^2 x_2)x_1 x_2 - x_2^2}{\theta_2}.$$

Integrating against $q = \mathcal{N}(0, I_2)$, only three numerical constants remain and the whole dependence on θ reads at a glance:

$$(g_{ij}(\theta)) = \begin{pmatrix} a & -\dfrac{\theta_1 d}{\theta_2} \\ -\dfrac{\theta_1 d}{\theta_2} & \dfrac{2+\theta_1^2 c}{\theta_2^2} \end{pmatrix}, \qquad \begin{aligned} a &= \mathbb{E}\big[\tanh^2 X\big] \approx 0{,}394, \\ d &= \mathbb{E}\big[X \tanh X\,(1-\tanh^2 X)\big] \approx 0{,}182, \\ c &= \mathbb{E}\big[(1-\tanh^2 X)^2 X^2\big] \approx 0{,}108, \end{aligned}$$

with $X \sim \mathcal{N}(0,1)$. Three things. The shear has a constant cost, g_{11} does not depend on θ: amplifying it costs the same price everywhere. The vertical dilation, for its part, has a hyperbolic cost in θ_2, exactly that of the standard deviation for the Gaussian, with an extra $\theta_1^2 c$ that makes it more expensive when a shear is already in place. Finally the cross term has the opposite

sign to the two movements requested here, θ_1 going up and θ_2 going down: leading them head-on costs more than separating them.

Figure 47 shows what this gives. The path stays very close to the log-linear interpolation in θ_2, within 10%: the vertical compression behaves like a change of scale and happens mostly at the beginning, when the shear is still weak. The densities start from a vertical ellipse, tilt, then curve. It is there that the non-affine character of the transport becomes visible: the tanh saturates, the points above and below the axis are shifted in opposite directions but by an amount that plateaus, and the iso-contours cease to be ellipses.

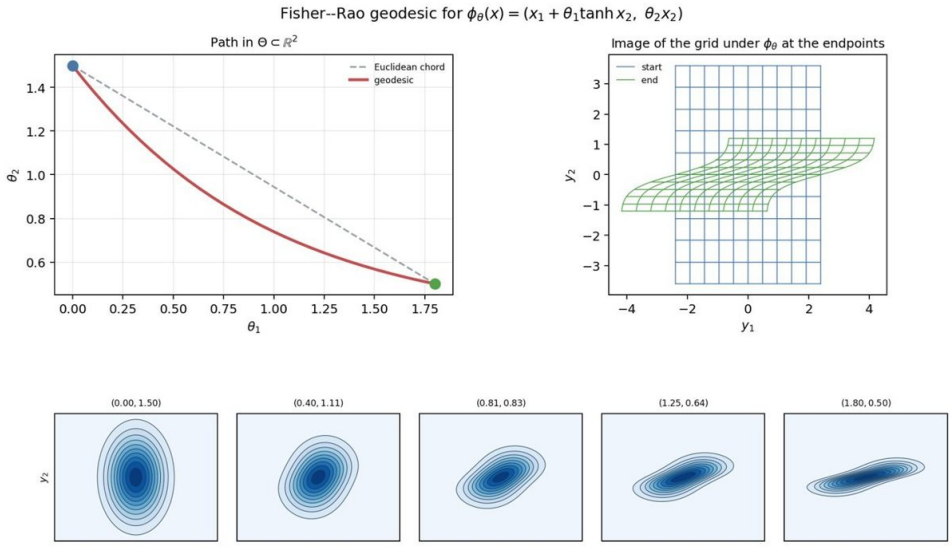


FIGURE 47. Numerical approximation of a Fisher–Rao geodesic for the two-dimensional transport $\phi_\theta(x) = (x_1 + \theta_1 \tanh x_2,\ \theta_2 x_2)$, base $q = \mathcal{N}(0, I_2)$. At the top left the path in Θ and the Euclidean chord; at the top right the image of a grid by ϕ_θ at the two endpoints, which shows the non-affine shear; at the bottom the densities along the path, with (θ_1, θ_2) in the title.

9. Numerical methods

Several figures display a path in parameter space. When the geodesic of g has a closed form, that formula is used directly. The remaining paths are obtained by minimizing a discrete Riemannian energy with fixed endpoints, in the spirit of variational discretizations of geodesic calculus [9]. We record here the functional, the solver, and the parameters figure by figure.

Let $\theta_0, \dots, \theta_N$ be a polygon in Θ, with $N = N_{\text{seg}} + 1$ and with θ_0, θ_N given. The N_{seg} interior vertices are the unknowns. The energy is

$$E(\theta_1, \dots, \theta_{N_{\text{seg}}}) = \sum_{k=0}^{N_{\text{seg}}} (\theta_{k+1} - \theta_k)^\top G\big(\tfrac{1}{2}(\theta_k + \theta_{k+1})\big)(\theta_{k+1} - \theta_k),$$

the metric being evaluated at the midpoint of each segment. This is a discrete version of $\int g(\theta', \theta')\, dt$, rather than of the length $\int \sqrt{g(\theta', \theta')}\, dt$: its minimizers are (approximate) constant-speed geodesics. The solver is L-BFGS-B. The tolerance in the table is a relative decrease of E. The initial polygon is the Euclidean chord, except for a log-linear interpolation of the σ_i when those coordinates are present, a small sinusoidal bump in (σ_1, σ_2) for figures 17–19, and a small transverse sinusoidal perturbation for figure 37.

figure	N_{seg}	max. iterations	tolerance	constraints
17–19	40	800	10^{-12}	$\sigma_i \in [0.2, 5]$, $\theta = \pi/4$ fixed
24, 25	36	600	10^{-11}	$\sigma_i \in [0.2, 5]$, $\theta \in [0, \pi]$
30 (Laplace)	36	800	10^{-10}	same
30 (Gauss)	36	600	10^{-11}	same
37	28	500	10^{-11}	none
46	28	600	10^{-12}	see below
47	28	800	10^{-13}	$\theta_1 \in [-4, 4]$, $\theta_2 \in [0.1, 4]$

For figure 46, the cubic family is constrained to $\theta \in [0.05, 2] \times [0.005, 1.2]$, the sinusoidal family to $\theta \in [0.01, 0.8] \times [0.2, 1.6]$.

The other geodesic figures are closed-form. Figure 14 is the Poincaré semicircle of the univariate Gaussian metric, sampled at 400 points. Figures 20–21 follow $\sigma_i(t) = \sigma_i(0)^{1-t}\sigma_i(1)^t$ at fixed μ and θ. Figures 22–23 are affine in θ at fixed μ and σ. In both of the last two cases the path is sampled at 200 points.

Figure 7 is different: the colouring is a numerical Riemannian distance on the graph of $\varphi(u, v) = (u, v, u^2 + v^2)$, obtained by geodesic shooting. Unit-speed geodesics are launched from $(1, 1)$ in 280 directions and integrated by a classical Runge–Kutta scheme of order 4, with 420 steps up to Riemannian time 3,2. Figure 1 does not compute geodesics: the two Gibbs panels show the Riemannian length of the Euclidean chord, by an 18-node quadrature of $\sqrt{\dot\theta^\top G(\theta(t))\,\dot\theta}$ along that segment.

10. Conclusion

The construction has been the same throughout the different families considered in this paper. A parametric model is represented by the map

$$\theta \longmapsto p_\theta \in \mathcal{P}_+(E),$$

and the parameter space is equipped with the metric obtained by pulling back the Fisher–Rao metric g^{FR} along this map. Whenever the parametrization is an immersion, this pullback defines a Riemannian metric on the parameter space. The geometry is therefore not imposed directly on the parameters: it is inherited from the way in which variations of θ deform the corresponding probability density.

At the infinitesimal level, the interpretation of Fisher–Rao has remained the same in all the examples. A small parameter displacement $d\theta \in T_\theta\Theta$

produces a density variation $dp_\theta(d\theta)$, whose squared cost is

$$g_\theta(d\theta, d\theta) = \int_E \left(\frac{dp_\theta(d\theta)(x)}{p_\theta(x)}\right)^2 p_\theta(x)\,dx.$$

Thus Fisher–Rao measures the relative variation of the density produced by the parameter displacement. In the interpretation developed throughout this paper, this corresponds to a local loss of overlap: the same absolute variation is more costly where the density is small than where most of the probability mass is concentrated. The pullback translates this infinitesimal cost into the coordinates of the parameter space.

The resulting geodesics, however, depend not only on the local expression of the metric, but also on the parametric family itself. A path in Θ gives a path of densities constrained to remain in

$$\{p_\theta : \theta \in \Theta\} \subset \mathcal{P}_+(E).$$

The shortest Fisher–Rao path between two densities in the ambient space $\mathcal{P}_+(E)$ may leave this subset and therefore need not correspond to an admissible path in the statistical model. The intrinsic geodesics considered here must instead follow the deformations allowed by the parametrization. In this sense, the geometry of each family reflects both the ambient Fisher–Rao metric and the particular way in which the family is immersed in the space of probability densities.

This common construction takes different forms in the families studied above. For the Gaussian family, the metric separates the effects of location and covariance. In the univariate case, it is, up to scale, the hyperbolic metric of the Poincaré half-plane: translating a sharply concentrated Gaussian is more costly than translating a diffuse one, and a geodesic may increase the variance before changing the mean. In higher dimension, mean displacements are measured through the Mahalanobis quadratic form, while the covariance block distinguishes changes of scale, anisotropy, and orientation.

The elliptical Laplace family exhibits a closely related decomposition. Location and orientation have the same geometric structure as in the Gaussian case, up to dimension-dependent factors, whereas the dispersion block contains an additional coupling between the principal scales. Similar endpoint configurations may therefore be connected through different intermediate deformations.

For a linear Gibbs family,

$$p_\theta(x) = \frac{1}{Z_\theta}\exp\bigl(-\phi(x)^\top\theta\bigr),$$

the metric takes the particularly rigid form

$$g_\theta = \mathrm{Cov}_\theta\bigl(\phi(X)\bigr) = \mathrm{Hess}_\theta \log Z_\theta.$$

The whole metric is thus generated by the single convex potential $\log Z_\theta$. Its anisotropy records the fluctuations of the features $\phi(X)$ under the current distribution, and degeneracy appears when some parameter direction produces an almost constant variation of the energy.

When the Gibbs energy depends nonlinearly on the parameters,

$$p_\theta(x) = \frac{1}{Z_\theta}e^{-U(\theta,x)},$$

the covariance interpretation remains:

$$g_\theta = \mathrm{Cov}_\theta\big(\nabla_\theta U(\theta, X)\big).$$

In general, however, the metric is no longer the Hessian of $\log Z_\theta$ alone. The additional term involving the curvature of U with respect to θ marks the loss of the Hessian structure specific to the linear family.

Finally, the push-forward construction gives a complementary interpretation. If

$$p_\theta = (\varphi_\theta)_\sharp q$$

and a parameter displacement $d\theta$ induces the velocity field $v_{d\theta}$ on E, then the corresponding density variation satisfies

$$dp_\theta(d\theta) = -\operatorname{div}_x\big(p_\theta v_{d\theta}\big).$$

Consequently,

$$\frac{dp_\theta(d\theta)}{p_\theta} = -\frac{\operatorname{div}_x\big(p_\theta v_{d\theta}\big)}{p_\theta}.$$

In this representation, Fisher–Rao does not measure the magnitude of the displacement field itself. What it sees is the local compression or dilation of probability mass generated by that field. A motion that rearranges points while leaving the density unchanged is therefore invisible to Fisher–Rao.

This last formulation also provides a natural bridge with Wasserstein–Otto geometry. The same continuity equation appears there, but it is used to define a different infinitesimal cost. Given a tangent density variation u, with

$$\int_E u\,dx = 0,$$

one considers all velocity fields v satisfying

$$u + \operatorname{div}_x(pv) = 0.$$

The Wasserstein–Otto norm is then

$$\|u\|^2_{W_2,p} = \inf_{\substack{v \\ u+\operatorname{div}_x(pv)=0}} \int_E \|v(x)\|^2 p(x)\,dx.$$

Under the usual assumptions, the minimizing velocity field can be taken to be a gradient field, and the associated Riemannian distance is the quadratic Wasserstein distance W_2 [8, 7].

The contrast with Fisher–Rao is particularly transparent at the infinitesimal level. For the same tangent density variation u, Fisher–Rao assigns the cost

$$\|u\|^2_{\mathrm{FR},p} = \int_E \frac{u(x)^2}{p(x)}\,dx,$$

whereas Wasserstein asks for the least kinetic energy required to realize u through a displacement of mass. Fisher–Rao measures the amplitude of a relative change of density; Wasserstein measures how mass must move through the geometry of the observation space. The former is insensitive to a transport that leaves the density unchanged, while the latter is built from the displacement field itself.

It would therefore be natural to repeat the construction of this atlas with the Wasserstein–Otto metric: pull it back along the same parametric families, derive the corresponding metrics on their parameter spaces, and

compare the resulting geodesics with those obtained here. Such a comparison would place the two geometries on exactly the same models and make explicit how statistical variation and spatial transport assign different costs to the same parameter displacement.

The purpose of the present atlas has been to keep the ambient Fisher–Rao geometry fixed and to vary the family through which it is observed. The examples show how this single infinitesimal notion of statistical discrepancy produces markedly different finite-dimensional geometries once pulled back to parameter space. The metric thus records how a model turns parameter variations into changes of probability law, while its geodesics describe the least costly statistical deformations that remain available within that model.

Code and figure availability

The code used to produce the numerical experiments and figures in this article, together with reusable versions of the figures, is available at https://github.com/PierreGuilbertF/pullback-fisher-rao-atlas. Unless otherwise stated, the figures are made available under the Creative Commons Attribution 4.0 International license (CC BY 4.0). When reusing the figures, citation of the present article is requested.

Statement on AI-Assisted Translation

This manuscript was originally written in French. Artificial intelligence tools were used to assist in translating the manuscript into English. The translation workflow was configured with author-defined instructions intended to preserve, as closely as possible, the tone, terminology, lexical choices, and style of mathematical exposition of the original French text.

The author has carefully reviewed and, where necessary, revised the complete English version. The author therefore takes full responsibility for the wording, accuracy, and scientific content of the English manuscript, notwithstanding the use of AI-assisted translation.

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